Endogenous Model Uncertainty and Curiosity under Cognitive Limits

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Abstract

What is curiosity: what leads a person to seek the information required to explore new perspectives? Neuroscience suggests that people automatically classify a situation before choosing how to act, drawing a familiar interpretation from similar past situations (e.g., a farmer attributing a neighbor's high yield to good weather). As observations become harder to reconcile with that interpretation, people increasingly entertain alternatives. This paper treats such shifts in "mindsets" as a choice of how much model uncertainty to entertain. Each mental model pairs a prior with a view of how observable features combine to reveal the state. After an initial signal, an agent with a costless default and latent, less precise alternatives jointly chooses model weights and how many subsequent signals to acquire. The key discipline is an evidence constraint that bounds motivated beliefs: an alternative's weight is capped by its Bayesian posterior. Curiosity then arises endogenously as the value of resolving the model uncertainty the agent chooses to entertain. Two results follow. First, the evidence constraint interacts with a complementarity between model weights and information acquisition to produce curiosity failures. Second, because agents can seek information and reason beyond what associative memory supplies, the framework isolates the influence of the information environment from hallmark behavioral effects in associative-memory models (e.g., Bordalo et al., 2025, 2026). Last, I endogenize the menu of mental models, formalizing creative and critical thinking as complementary forms of cognitive dimension reduction that sustain curiosity. I illustrate the theory throughout with development programs, teacher training, and agricultural technology adoption.

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A mind is not a vessel to be filled, but a fire to be kindled.

— Plutarch, *On Listening to Lectures*

The efficiency of a human problem solver depends critically on his ability to find a representation of the problem that restricts search to a small subspace of the total problem space.

— Allen Newell and Herbert A. Simon, *Human Problem Solving* (1972)

1 Introduction

Is seeing a doctor a path to better health, or a risk of the mistreatment once inflicted on people of one's race (Alsan and Wanamaker, 2018)? Is a neighbor's high yield the result of new farming practices, or simply of favorable rainfall? Is school-based agricultural education in rural Africa a way to build students' independent farming skills and scientific capabilities while spreading improved techniques to their households, or does it simply train students as farm labor at the expense of their studies, as many school gardens do (Lee, 2026)?

In each of these cases, how a person classifies the situation shapes her behavior decisively. Each classification tells her not only what to expect but where to look: a farmer who credits the rainfall has no reason to compare her neighbor's plots with her own. Importantly, when past experience leads a person to dismiss a promising opportunity, providing facts has limited effect, because whether her reading is correct becomes clear only through personal, repeated observations. Medical care unfolds over repeated visits, the source of a neighbor's yield emerges only from comparisons across plots and seasons, and the gains from school-based agricultural education become clear only through parents' conscious efforts to monitor its progress: visiting school farms, assigning plots for student home projects, and attending meetings with teachers, which few rural parents do regularly. Gathering this evidence takes time and resources, and keeping track of an unfamiliar reading takes mental effort. When past experience makes people confident in their familiar interpretation, such evidence promises too little to justify these costs. Consequently, the focal question becomes how to stimulate curiosity—the drive for people to seek, on their own, the information required to explore new perspectives.

In fact, these examples reflect a general feature of cognition. A large body of neuroscience, most prominently cognitive map theory (O'Keefe and Dostrovsky, 1971; O'Keefe and Nadel, 1978; Hafting et al., 2005), suggests that people do not weigh every interpretation of a situation afresh. To economize on mental effort under cognitive limits, the brain instead assigns each situation to a familiar mental model[1] retrieved from memory of similar past experience, before deliberation begins (Behrens et al., 2018; Sanders et al., 2020). As observations become harder to reconcile with that model, it increasingly entertains alternative mental models, but there is also room for choice: the brain tends to favor mental models that promise higher payoffs if true and ignore those that are plausible but less rewarding (Shenhav et al., 2013; Lieder and Griffiths, 2020).

Allowing people to seek information and reason beyond what associative memory supplies, this paper models these contextual shifts in "mindsets" as a choice of how much model uncertainty to entertain after an initial signal—such as how a development program is presented. These "mindsets", which are competing models of the world formally known as mental models, are defined as a prior over the payoff-relevant state together with a likelihood that says how observable features combine to reveal that state. Two mental models

[1] I use the terms "mental models" and "cognitive maps" interchangeably. Neuroscientists speak of cognitive maps because the concept grew out of studies of how the hippocampus represents physical space, and was later extended to the organization of abstract knowledge (O'Keefe and Nadel, 1978; Behrens et al., 2018); psychologists use "mental models" for the internal representations people reason with. This paper relies only on what the two share: a structured representation that tells a person which observable features matter and how they combine to reveal an unobserved state. It makes no claim about how that representation is implemented in the brain, so the two terms can be used interchangeably here.

therefore differ not only in what they expect but in what they attend to and how they read it, and what each attends to follows from its likelihood rather than being assumed. The agent is not restricted to holding only one mental model at a time, and weighting mental models changes how a person interprets a situation.[2] The agent starts from a default mental model that is costless to use and faces a menu of latent alternatives[3] whose priors are less precise than the default's and that yield expected payoffs that differ from those under the default. After observing the initial signal, she jointly chooses (i) the weight of attention she pays to each alternative and (ii) how many signals to acquire.

Regarding the choice of mental model weights, the framework integrates two decision theories under cognitive realism. As in Bayesian model averaging (e.g., Brock, Durlauf, and West, 2003; Cogley and Sargent, 2005), the posterior disciplines the weights, but the agent chooses them rather than setting them equal to it: tracking an unfamiliar reading takes mental effort, and holding it forces a single action to hedge across models that recommend different actions, so she may place less weight on an alternative than its posterior. As in models of motivated beliefs (e.g., Brunnermeier and Parker, 2005; Bénabou and Tirole, 2016), expected payoffs guide that choice. However, I impose a key discipline from neuroscience: the brain activates a mental model according to how well it predicts what is observed relative to its rivals, so a mental model the evidence does not support cannot be activated, however attractive it is.[4] This evidence constraint caps the weight on an alternative at the posterior probability that the evidence alone assigns it: the agent can neglect evidence but cannot invent it.

In summary, the agent places weight on an alternative to the extent that acting on it promises, if it is right, a higher expected payoff than acting on the default, net of the costs of holding it, subject to the evidence cap. The evidence cap plays an analogous role to collateral constraints in models of poverty traps (e.g., Galor and Zeira, 1993; Banerjee and Newman, 1993), generating threshold effects in the exploration of alternative mental models. To see why, consider an alternative that is more promising than the default—less promising alternatives are never activated. If the alternative is right and evidence kept arriving, the cap would rise steadily and the agent would shift her weight gradually toward the alternative. Two features of the setup break this convergence. First, the initial signal fixes the evidence cap. Second, model weights and information are complements: information is worth more under the vaguer alternative than under the precise default, so the more weight she places on the alternative, the more she seeks. Together, these features create a threshold. A weak initial signal caps the weight below the level at which entertaining the alternative pays, so the agent never investigates it, the cap never rises, and the alternative stays unexplored—a *curiosity failure* in which she never seeks the information that could change her mind. A signal just above that level starts a self-reinforcing cycle: the weight justifies information, the information raises the cap, and the higher cap justifies more weight. Small differences in the initial signal can thus produce sharply different outcomes. Parents in Africa may therefore ignore a school-based agricultural education program if it fails to distinguish itself from typical school gardens that use junior high students as farm labor. Yet a light-touch information intervention can prompt them to seek information on their own, through school visits, trials of improved techniques, and delegating farms to students (Lee, 2026).

Given the central role of the initial signal, it is natural to ask which signals clear the threshold. The

[2] When the context is ambiguous, the brain does not commit to a single mental model (Sanders et al., 2020). Instead, it flickers between familiar and new mental models and searches for evidence until it settles on one. This flickering lasts fractions of a second in spatial navigation (Jezek et al., 2011), longer in the perception of ambiguous figures (Leopold and Logothetis, 1999), and plausibly months for the conceptual mental models studied here. A decision taken after deliberation integrates the mental models the brain has visited.

[3] A latent alternative need not come from experience; it is available but carries no weight until activated. Latent structure of this kind has a neural counterpart: sequences that later form the mental model of a new environment are already present in the hippocampus before that environment is first experienced (Dragoi and Tonegawa, 2011). Sections 5 and 6 discuss in detail how mental models are formed from templates and how the resulting menu is pruned.

[4] In the hippocampus, this matching works through attractor dynamics: incoming cues pull activity toward whichever stored mental model they fit best (Wills et al., 2005). The resulting assignment behaves like Bayesian inference over which context applies (Sanders et al., 2020), which is why the constraint takes the form of a posterior.

answer is not simply those that supply more observable features that fit the alternative. Unlike in models with one-dimensional signals, how evidence is framed is central here, for two reasons: the evidence cap rests on relative likelihoods, and a signal is a vector of observable features that each mental model reads through its own attention. The signal can raise the evidence caps on three margins. The first is surprise: the alternative is less surprised by the part of the signal it reads than the default is by the part it reads. The second is encoding: the signal carries features that the alternative reads but the default ignores. The third is specificity: the closer the alternative's prior comes to the default's in precision, the smaller the penalty it pays for spreading its predictions thin. Two implications follow. First, exploration can be stimulated while holding constant the observable features the default reads, by presenting unique features an alternative reads or by stating the alternative as a specific, checkable claim consistent with what is observed. Second, a signal can fail to change behavior not for lack of observable features but for lack of any mental model that reads or explains them.

Characterizing the choice of model uncertainty then allows curiosity to be defined formally, as an endogenous object rather than as a fixed trait. Once an alternative mental model carries weight, the decision maker must act without knowing which mental model is right, which forces her to hedge and makes resolving the differences between perspectives valuable. Curiosity is defined exactly as the value of resolving that model uncertainty, which is a function of model weights. It is zero under the default alone and greatest when the mental models are most balanced, falling away as either prevails. This inverted-U relationship matches the well-known finding that curiosity peaks at intermediate confidence (Loewenstein, 1994; Kang et al., 2009). The framework can further integrate non-instrumental motives for curiosity by adding ambiguity aversion, applying a concave aggregator to the weights she chooses. Ambiguity aversion raises both the cost of holding an alternative and the value of resolving which mental model is right, and the ambiguity premium grows with the gap in expected payoffs between the mental models. Questions with high stakes—such as whether fertilizer bought on credit will double a farm's income or, if it is fake, leave the family in debt (Bold et al., 2017; Michelson et al., 2021)—are therefore harder to open but, once open, more actively pursued. In such settings, interventions that activate alternative mental models become more effective relative to those that make information cheaper to acquire.

Because each mental model reads observations through its own lens, not every observation can tell the mental models apart, which raises the question of where information is sought. I thus extend the model so that the decision maker chooses not only how many signals to acquire but also the type of signals, so that the information she is exposed to is endogenous. The type of signal she chooses trades off learning more within the mental models she holds (refinement) against learning which of them is right (resolution), and the weight she places on the alternative sets the terms. With little model uncertainty, she seeks few signals, since a precise default has little left to learn, and those she seeks are ones the default reads. As model uncertainty grows, curiosity directs her toward signals about the features on which the mental models disagree, most strongly when the answer would change what she does. Once the alternative dominates, she seeks signals the alternative reads. The shifts in "mindsets" are thus reflected in what information the decision maker seeks: a farmer who credits the rainfall watches the weather; one who suspects a new practice compares plots; one who has adopted it studies how to apply it.

An essential feature of my framework is that it grants decision makers the capacity to reason and explore beyond what associative memory supplies. This revises two hallmark behavioral effects in associative-memory models. The first is context effects. In these models, adding payoff-irrelevant cues can change which representation of the situation comes to mind and change behavior accordingly (Bordalo et al., 2026). For example, a consumer sees a jar of jam in plain packaging as a routine purchase and focuses on its price, but the same jar in fancy packaging becomes a treat and draws her attention to its quality. In my framework, a cue is defined as an observable feature of the initial signal and counts as evidence in the same way as any other feature. Each mental model assesses how probable the whole signal, cue included, would be if it were right, and the cue shifts weight only toward a mental model that predicts it better than the others. Cues therefore sway behavior only if they are payoff-relevant under at least some mental models. Moreover, when a cue

activates an alternative that reads it, the endogenous curiosity this triggers tests that alternative: with unbiased evidence, the shift persists if the alternative fits the evidence better than the default and reverses otherwise. When one of the two mental models is true, departures from optimal search, such as a tendency to seek confirming evidence, only slow the reversal. Persistent context effects therefore point to evidence that is scarce, costly, or biased, or to a menu with no true mental model, rather than to a fixed cognitive bias. Because people have an endogenous drive to seek this evidence themselves, policy can make it accessible instead of delivering it: for farmers who distrust fertilizer that looks clumped but meets nutrient standards (Michelson et al., 2021), the remedy is quality testing they can obtain on their own.

Thus far, the discussion has pitted a single alternative against the default, but practical questions often admit many explanations: a farmer who sees a neighbor's high yield may credit the weather, the soil, a new seed variety, fertilizer, or the timing of planting. My framework allows many alternatives, and because the evidence cap counts all of them, the support the evidence lends one shrinks with every rival that also fits it. This evidence-based reasoning means that interference effects, the second hallmark, also work differently. In associative-memory models, different experiences compete for retrieval (Bordalo et al., 2025), but the representation with the greatest cue similarity is the one that comes to mind. In my framework, an alternative is explored only if the evidence gives it enough support, and rivals that fit the same evidence divide that support. Several alternatives that each outperform the default can therefore all fail to be activated. Even a contextual cue that lends plausibility to an alternative less promising than the default—and thus one the decision maker never entertains—can stifle exploration of a promising one.

Interference effects thus raise a question that a fixed menu of mental models cannot answer: how does a mind with limited capacity keep its menu small enough to explore at all? I next endogenize the menu and introduce creative and critical thinking as two complementary forms of *cognitive dimension reduction*. Both rest on a premise from neuroscience: mental models are assembled from reusable pieces of explanation—formally known as templates—each linking an underlying cause to the observable features it produces, such as rain to wet soil and high yields (Behrens et al., 2018; Whittington et al., 2020), and the brain imagines new situations by recombining stored experience (Hassabis et al., 2007; Schacter and Addis, 2007).[5] Formally, a template is a causal mechanism, and a mental model is the causal graph assembled from such mechanisms (Pearl, 2009). In the linear Gaussian case used here, a template is a latent factor: a loading vector that records which observable features the cause moves, together with a prior on how strongly it acts and on how far that strength is invariant across situations. As with curiosity, I model creative thinking as an evidence-based process: the agent builds new mental models from templates she already holds, and each step must be justified by the evidence. I distinguish two forms. *Vertical* creative thinking adds a template to an existing mental model, explaining the same situation more deeply, such as crediting a neighbor's yield to the soil as well as the rain; I show that it is justified exactly when what the current mental model fails to predict is surprising enough along the new template's direction, where the surprise required rises with the template's vagueness. *Horizontal* creative thinking extends a mental model to new situations, such as carrying a lesson learned on cassava over to rice; it is justified, I show, when the situations depart from expectations in the same direction. I prove that the two forms interact: a vertical step that adds a template explaining why a technique works lets the mental model transfer further horizontally. Creativity is thus not a fixed trait: it is bounded by the templates the agent holds, so providing templates expands what she can create and how far it transfers. This matters for training teachers and agricultural extension officers: a set protocol gives them a mental model precise enough to be activated where it was designed, whereas understanding why a practice works supplies a template that lets them adapt it to classrooms and farms the protocol never anticipated, a gain that grows

[5] Recent work suggests what a reusable template looks like in the brain. Neural populations in the hippocampus and prefrontal cortex organize what they encode on low-dimensional surfaces, or manifolds, with a shared geometry, so that a pattern learned in one setting carries over to new ones (Bernardi et al., 2020), much as a template carries over to new situations. The hippocampus also represents space on a hyperbolic manifold that expands as experience accumulates (Zhang et al., 2023), so the representation grows with what has been learned.

with the number of situations it covers but depends on their own creativity and curiosity to build and test each adaptation.

Creative thinking has a potential pitfall, however: a mental model assembled after seeing the evidence can fit it by chance. Critical thinking, a standing habit of mind, guards against this. Informally, it means asking whether a convincing story would have fit whatever happened, and checking it on cases not used to find it. More precisely, it means not using the same data multiple times: every mental model, new or old, is judged only by evidence not used to form it. I model it as a parameter $\kappa \in [0,1]$, how strongly the agent discounts an explanation found after the fact for the breadth of the search through her templates that produced it. The discount applies to any mental model selected after seeing the evidence it is judged on. The structure is that of multiple hypothesis testing: each candidate explanation is a test, and a false insight is a false positive. I prove that a threshold value of κ, strictly below one, separates two regimes. Below it, the probability of a false insight converges to one as the number of candidates searched grows; above it, that probability converges to zero, because the evidence required for admission rises faster than the best fit that chance alone produces. Critical thinking thus prunes the menu, keeping false insights from diluting the evidence cap of the right alternative. Critical thinking also comes with its own potential pitfall: it makes a correct explanation harder to admit, and more so the broader the search that found it. Without complementary triggers for creative thinking, school-based agricultural education programs that cultivate critical thinking face a tradeoff between fewer false insights and slower adoption of correct ones: students trained to think critically should be less likely to adopt a new planting date or seed variety that merely coincided with a good harvest, but they should also be slower to adopt effective crop spacing they discover themselves.

Taken together, the results suggest that open-mindedness, curiosity, creative thinking, and critical thinking form a cycle within the brain's evidence-based reasoning. Open-mindedness, entertaining an alternative as far as the evidence allows, starts curiosity, but curiosity tends to extinguish itself as the evidence it gathers resolves the uncertainty that prompted it. Creative thinking can replenish that uncertainty, since the surprises curiosity uncovers justify new mental models, while critical thinking judges each against fresh evidence and keeps chance fits off the menu, where they would dilute the evidence cap of the right alternative. These capabilities are often treated as general traits or skills that a person carries across settings (McCrae, 1987; Halpern, 1998; Heckman and Kautz, 2012; Kashdan et al., 2018). Viewed through the brain's evidence-based reasoning, however, even a general capability must start somewhere: the evidence of a particular situation sets it in motion, and carrying it to new situations is a creative step that the evidence must justify and that costs mental effort. This offers one reading of why efforts to train these capabilities in the abstract have shown limited transfer across domains (Willingham, 2007; Sala and Gobet, 2017). The same problem arises for training teachers in inquiry-based instruction, which they need to tailor to diverse contexts. How, then, can they learn to do so and keep growing their capabilities? The framework suggests an integrative approach that keeps the cycle going: sharpening inquiry-based templates at a more fundamental level rather than giving out recipes, while providing streams of information that stimulate teachers to create their own mental models, using mental models with lower but more certain payoffs as stepping stones to ones with higher but less certain payoffs. Section 7 discusses my ongoing research that tests this idea, integrating a highly effective "Learning to Learn" approach to teacher training and community transformation (Nourani et al., 2025, 2026) with school-based agricultural education as a platform that offers experiential science learning opportunities and innovative farming techniques (Lee, 2026).

Organization. Section 2 discusses the model setup and key assumptions. Section 3 characterizes alternative activation, curiosity failures, what signals clear the threshold, a formal definition of curiosity, and where the decision maker seeks information. Section 4 illustrates how behavioral effects are revised relative to associative-memory models. Sections 5 and 6 endogenize the menu of mental models through creative and critical thinking. Section 7 offers concluding remarks.

2 Setup

An agent faces an unknown payoff-relevant state and holds a menu of mental models of her situation: a costless default and K latent alternatives. After an initial observation, she chooses jointly (i) the attention weights she places on the alternatives, which set how much model uncertainty she entertains, and (ii) how many costly signals to acquire, that is, "active" learning. She then takes a single act that must serve every model she holds. Two limits discipline the weights: holding alternatives is cognitively costly, and no alternative may receive more weight than the evidence supports. The agent maximizes expected utility; ambiguity aversion enters only as an extension, where it adds a non-instrumental motive for curiosity and strengthens the results. The menu is taken as fixed throughout. The structure rests on three insights from neuroscience, specifically cognitive map theory, which originally dealt with spatial navigation but has recently been extended to learning about conceptual ideas, memory mapping, and social relationships (Constantinescu et al., 2016; Behrens et al., 2018; Whittington et al., 2020):

- **Default, familiar preconception (auto-pilot mode).** Agents begin with a costless default prior that is more precise than the alternatives (Kool et al., 2010; Shenhav et al., 2013).
- **Costly activation of latent mental models.** Activating an alternative prior requires cognitive effort, modeled as a convex cost, and holding multiple alternatives at once is more costly than holding each alone (Shenhav et al., 2017; Lieder and Griffiths, 2020).
- **Complementarity between activation and information acquisition.** The more an agent commits to a new prior, the more valuable information about it becomes; conversely, such information raises the value of the new prior.

2.1 Mental models, signals, and timing

Let $\theta \in \mathbb{R}$ denote an unknown payoff-relevant state: the return to a course of action whose value the agent does not know. The agent has von Neumann–Morgenstern preferences represented by $u(a,\theta)$ over acts $a \in \mathscr{A} \subseteq \mathbb{R}$ and states θ, and $\mathbb{E}_\mu\, u(a,\theta)$ denotes expected utility under a belief μ.

Mental models. The agent holds a finite menu of mental models of the situation she faces, indexed by $k \in \{0,1,\ldots,K\}$. Index $k=0$ is the *default* model, which she has applied many times before and which she operates on costlessly; indices $k=1,\ldots,K$ are *latent alternatives*, which she may or may not bring to bear. The menu is fixed: subsequent signals reweight and refine the models on it rather than introducing new ones. Taking it as given is where an associative account of memory enters: what an agent can entertain is what the situation has brought to mind, so retrieval determines the menu and the problem studied here begins once retrieval has happened. Nothing below requires the menu to be the same across situations or agents, only that the weight placed on whatever is available be disciplined by the evidence. A mental model has two aspects. The first is a *prior* over the state: model k asserts $\theta \sim \mathcal{N}(\mu_k, \sigma_k^2)$ with precision $\tau_k \equiv 1/\sigma_k^2$, and the default is the most precise on the menu, $\tau_0 > \tau_k$ for every $k \geq 1$, because it has been applied most often. The means are left unordered. The second aspect is a *generative process* that says how the observable features combine to reveal the state, which determines both what the model attends to and how it interprets what it sees. The agent observes a raw feature vector $x \in \mathbb{R}^d$, such as the features of a message or what is seen on a visit, with the scalar case recovered at $d=1$. The vector includes contextual features, such as where, when, or by whom a situation is presented, alongside features that bear on payoffs under the default. A contextual cue is thus a component of x like any other: it is evidence only under a mental model that reads it. As the cognitive-map theory treats it, each model is a complete generative process for those features:

$$\theta \sim \mathcal{N}(\mu_k, \sigma_k^2), \qquad x \mid \theta \sim \mathcal{N}(\theta\, b_k,\, \Sigma), \qquad \tau_k := 1/\sigma_k^2, \qquad k = 0,1,\ldots,K, \tag{1}$$

where $b_k \in \mathbb{R}^d \setminus \{0\}$ is the mental model's loading vector and $\Sigma \succ 0$ is a common noise covariance.[6] Each alternative is structurally distinct from the default and from the other alternatives, in that it carries its own loading b_k.

The attention mapping is not an extra primitive. It is what each mental model's own inference problem implies.

Lemma 1 (Attention mappings are derived)**.** *Under* (1)*, the posterior over θ given x depends on x only through the scalar*

$$s_k \;=\; \alpha_k' x, \qquad \alpha_k \;=\; \frac{\Sigma^{-1} b_k}{b_k' \Sigma^{-1} b_k}, \tag{2}$$

and $s_k \mid \theta \sim \mathcal{N}(\theta, \nu_k^2)$ with $\nu_k^2 = (b_k' \Sigma^{-1} b_k)^{-1}$. The posterior is Gaussian with precision $P_k(0) = \tau_k + \nu_k^{-2}$ and mean $m_k = \big(\tau_k \mu_k + \nu_k^{-2} s_k\big)/P_k(0)$.

Each mental model thus compresses the observable features into a single number s_k carrying everything it needs to know about θ, and the weights in that compression, α_k, are its attention mapping, the combination of features its own model treats as informative. When $b_j \neq b_k$ two models read the same raw observation differently, and features diagnostic under one may be ignored by the other, so shifting weight between them changes how the agent interprets the same situation; when $b_j \propto b_k$ they attend identically and differ only in (μ_k, τ_k). The distinction has observable content, since models with different loadings assign different likelihoods to the same features. This gives a formal edge to what is otherwise verbal: two priors differing only in their means are *not* two mental models in any sense that matters here.

Signals. After observing x, the agent may acquire $n \geq 0$ subsequent signals at marginal cost c per signal, each of the form $y_i = \theta + \eta_i$, $\eta_i \sim \mathcal{N}(0, \nu_y^2)$.

Attention weights. The agent chooses a vector of attention weights $\mathbf{w} = (w_1, \ldots, w_K)$ on the alternatives, with $w_k \geq 0$ and $\sum_{k \geq 1} w_k \leq 1$, the remainder $w_0 = 1 - \sum_{k \geq 1} w_k$ resting on the default; and she chooses the number of signals n. Timing is:

(1) x is realized and observed.

(2) The agent chooses the weights $w_k \in [0, \overline{W}_k(x)]$ on the alternatives and the number of costly signals $n \in \{0, 1, 2, \ldots\}$.

(3) The n signals realize.

(4) The agent chooses a single act a, evaluated at the weights chosen in step (2).

The results concern the weights chosen at step (2). Step (4) fixes the structure of the problem. The agent chooses a single act under model uncertainty, so the maximization runs over one a evaluated at the aggregated weights. Aggregating model-contingent maxima $\max_a \mathbb{E}_k u(a, \theta)$ would instead evaluate the position of an agent who *knows* which mental model applies. Placing the maximum outside the sum over models is what gives the disagreement between mental models its bite, and it generates the hedging cost defined below.

The model is a single choice followed by a Bayesian update: the agent chooses $(\mathbf{w}, n)$ once, the signals realize, the weights update, and she acts. Statements about the path of a shift, such as a shift that carries itself forward or rival alternatives eliminated one at a time, apply to repeating this stage, with the updated evidence caps carried forward as the bounds of the next stage and the agent choosing myopically in each. Myopia is consistent with the neuroscience: the brain reassigns a situation to a mental model as evidence arrives, through ongoing inference about which context applies, rather than by planning future reassignments (Sanders et al., 2020).

[6] Allowing Σ_k to differ across mental models changes nothing conceptually and adds a fourth channel to the decomposition of the Bayes factor. A common Σ isolates the channels of interest.

Updating. Write $\Lambda_k^y(y^n)$ for the likelihood ratio of the n acquired signals under alternative k against the default, each mental model evaluated at its own step-(2) posterior, so that the predictive probability model k assigned to the signals that were realized is what enters. After the signals the weights and the evidence caps update by Bayes' rule,

$$w_{k,n} \;=\; \frac{w_k\,\Lambda_k^y(y^n)}{w_0+\sum_{j=1}^K w_j\,\Lambda_j^y(y^n)}, \qquad \overline{W}_{k,n} \;=\; \frac{\Lambda_k(x)\,\Lambda_k^y(y^n)}{1+\sum_{j=1}^K \Lambda_j(x)\,\Lambda_j^y(y^n)}, \tag{3}$$

and these carry forward to the next stage. This is the recursive update of the second-order prior in Klibanoff et al. (2009), applied to weights the agent has chosen rather than inherited; the within-model posteriors update as before. When a single alternative k is held, (3) reduces to $w_{k,n} = w_k\Lambda_k^y/(1-w_k+w_k\Lambda_k^y)$. Updating preserves the discipline: the odds of every alternative against the default never exceed its Bayes factor, before or after the signals, and the weights are a martingale under the mixture, so acquiring signals cannot be expected to move attention in either direction, only to resolve it (Lemma OA.3 in the Online Appendix).[7]

2.2 Preferences, costs, and the derived objective

The agent evaluates an act by its expected utility under each mental model and aggregates across models with the weights. To nest attitudes toward model uncertainty, the aggregation is written through a strictly increasing $\varphi:\mathbb{R}\to\mathbb{R}$, as in the smooth ambiguity model (Klibanoff et al., 2005), except that the weights are chosen rather than given by preferences. The baseline sets $\varphi(U)=U$: the agent maximizes expected utility under the mixture of models and has no ambiguity aversion as such. A strictly concave φ adds ambiguity aversion. It raises both the cost of holding an alternative, through higher activation thresholds, and the value of resolving which mental model is right, by directing information seeking toward evidence that discriminates between models; both effects grow with the gap in expected payoffs between the mental models. It strengthens the results rather than drives them.

Holding alternatives active imposes a cognitive cost,

$$C(\mathbf{w}) \;=\; \frac{\gamma}{2}\sum_{k\geq 1} w_k^2 \;+\; \eta \sum_{1\leq j<k\leq K} w_j w_k, \qquad \eta>\gamma>0. \tag{4}$$

The cost has three properties. First, operating under the default alone is costless. Second, it is convex: each additional unit of attention to an alternative requires progressively greater inhibition of the dominant default and progressively greater suppression of interference from other active alternatives.[8] Third, because $\eta>\gamma$ the cost is superadditive across alternatives: holding two alternatives at once costs strictly more than the sum of holding each alone. When a single alternative k is held, (4) is simply $(\gamma/2)w_k^2$. A count-based cost such as $\gamma(m-1)^2$ in the number m of active models, by contrast, makes the problem a mixed integer program with no first-order conditions in the weights. The agent weighs the value function against this cost and the information-acquisition cost cn, selecting $\mathbf{w}$ and n to maximize the objective below subject to the evidence constraints $0\leq w_k\leq \overline{W}_k(x)$, defined below, and $n\in\mathbb{Z}_{\geq 0}$.

The cost (4) is not an information cost. A relative-entropy penalty for departing from a reference distribution over models, as in robustness and rational inattention (Hansen and Sargent, 2008; Sims, 2003), *rewards*

[7] Within a stage the weights move only by Bayes' rule, and the agent anticipates them correctly, since they are a martingale under the mixture; re-choosing them is deliberation and begins the next stage. Given the weights, she maximizes expected utility and updates the mixture by Bayes' rule, so her choice of act is dynamically consistent. Activation is a choice of how to deploy deliberation, not a preference over acts.

[8] The convex form is the natural reduced-form summary of two well-documented neural constraints. Working-memory capacity is severely limited, to roughly three to four items (Cowan, 2001), and the precision with which items are held drops nonlinearly rather than linearly as memory load increases (Bays and Husain, 2008). And tracking competing explanations is more demanding than tracking items, because each hypothesis carries its own predictions that must be updated against incoming evidence.

spreading weight across alternatives, since entropy favors dispersion. The interference term $\eta > \gamma$ does the opposite, and it is what the working-memory capacity limit contributes (Cowan, 2001; Bays and Husain, 2008). It gives the agent a preference for entertaining one alternative at a time.

Writing $\mathbb{E}_{k,n}$ for the expectation under mental model k's posterior after x and n signals, and $\mathbb{E}^y$ for the expectation over signal realizations taken at step (2), the derived objective is

$$V(\mathbf{w}, n; x) \;=\; \mathbb{E}^y\Big[\max_{a\in\mathscr{A}} \sum_{k=0}^{K} w_k\, \varphi\big(\mathbb{E}_{k,n} u(a,\theta)\big)\Big] \;-\; C(\mathbf{w}) \;-\; c\,n. \tag{5}$$

Equation (5) is built from vNM utility, the aggregation across models, and convex costs. It is a behavioral model assembled from these primitives rather than a representation of preferences over acts: the weights are a choice variable, not objects revealed by choice as the second-order prior is in Klibanoff et al. (2005). The agent is not maximizing a cardinal value function for its own sake; she is choosing how much model uncertainty to entertain, jointly through the weights $\mathbf{w}$ and the signal count n, subject to the cognitive cost and the evidence caps. Nothing in (5) is primitive except u, φ, γ, η, and c.[9]

2.3 The evidence constraint

The evidence constraint means that the weight on an alternative cannot exceed the evidentiary support for that alternative: it is a constraint on likelihood, not on value, and it binds what the agent may believe rather than what she may observe. It is cognitive in origin even though its content is evidentiary, because it is what the brain's own assignment of a situation to a model leaves for deliberation; the cognitive cost (4) is the second limit, and the two together are the cognitive limits of the title. This has a neural basis: categorical activation is justified by evidence, not by preference; an agent cannot adopt a prior the data do not support, no matter how attractive its mean is (Sanders et al., 2020). Let $L_k(x)$ be the probability that mental model k assigned, before the fact, to the features that were actually observed, its marginal likelihood, and let $\Lambda_k(x) := L_k(x)/L_0(x)$ be the Bayes factor of alternative k against the default: how much better alternative k accounts for what was seen than the default does. Define

$$\overline{W}_k(x) \;:=\; \frac{L_k(x)}{L_0(x)+\sum_{j=1}^{K} L_j(x)} \;=\; \frac{\Lambda_k(x)}{1+\sum_{j=1}^{K}\Lambda_j(x)}, \qquad k=1,\dots,K. \tag{6}$$

When x is consistent with the default, L_0 dominates and every $\overline{W}_k \approx 0$, foreclosing meaningful weight on any alternative. When x falls in the tails of the default, the ratios reverse and the bounds of the alternatives that account for x rise. The right-hand side of (6) is exactly the posterior probability a Bayesian with a uniform prior over the $K+1$ mental models would assign to alternative k after seeing x, so the constraint $w_k \le \overline{W}_k(x)$ reads: *the agent may not be more convinced of an alternative than the evidence justifies*. The constraint is one-sided. She may remain at $\mathbf{w}=0$ however strong the evidence, and does so whenever an alternative promises too little to pay for holding it, which is the sense in which the default is sticky; she may never exceed $\overline{W}_k$, which is the sense in which the model is not one of motivated beliefs. A default-favoring prior over mental models rescales L_0 in (6): it lowers every bound and so raises the thresholds below, but none of the results changes in form. Two features of the bound matter for what follows. The denominator includes every alternative, not just the focal one,[10] so each alternative's admissible weight depends on the evidentiary

[9] What the analysis uses is an explicit second-order belief over mental models that updates by Bayes' rule (3). The main results set φ linear, so they hold for an expected-utility agent; the curvature of φ matters only in the extension to ambiguity aversion. Representations without a second-order belief, such as maxmin (Gilboa and Schmeidler, 1989), variational or confidence preferences, have no weight to cap and no counterpart to activation.

[10] The cap requires no deliberate computation. It is set by the automatic assignment described below, which matches cues against all stored mental models at no cognitive cost, as hippocampal attractor dynamics do (Wills et al., 2005; Sanders et al., 2020); the cognitive cost (4) applies only to the mental models deliberation holds. Latent mental models exist before they are used (Dragoi and

support enjoyed by every other alternative the agent might entertain, so rival explanations dilute one another's support. And when only one alternative is on the menu the bound is $\Lambda_k/(1+\Lambda_k)$, the posterior probability in a two-model comparison.

Where the constraint comes from. The bound has a natural counterpart in the brain's automatic assignment of a situation to a mental model: it is what that assignment leaves for deliberation. Sanders et al. (2020) show that hippocampal remapping behaves like Bayesian inference over which context the animal is in: the mental model that becomes active is the one favored by the posterior over contexts given the evidence. Taking that as the automatic process, after x it assigns to model k

$$p_k(x) \;=\; \frac{q_k L_k(x)}{\sum_{j=0}^{K} q_j L_j(x)}, \tag{7}$$

where q is the prior the automatic process carries. Deliberation can hold a model only to the extent the automatic process has made it available, $w_k \leq p_k(x)$; with a uniform q this is (6), and a default-favoring q rescales L_0. The one-sidedness has its own counterpart: a well-learned mental model is an attractor, so remaining in it requires no evidence and it snaps back after a brief disturbance (Wills et al., 2005; Jezek et al., 2011), whereas leaving it requires evidence. This is a reading of the neuroscience rather than a derivation from neural dynamics; what it commits the model to is the property the results use, that deliberate activation is bounded by posterior probability.

Two limiting cases. The constraint places the agent between two benchmarks the literature treats separately. Were holding models costless, she would set every weight at its cap: that is Bayesian model averaging, in which weights are posterior probabilities and nothing is chosen. Were there no cap, she would set the weights by the value of the models they favor, tempered only by the costs: that is the structure of optimal expectations (Brunnermeier and Parker, 2005), which Proposition OA.2 in the Online Appendix states exactly. The agent lives on the segment between them: her weights are chosen, which the averager's are not, and bounded by evidence, which the optimist's are not. Everything the paper adds (activation as a choice, curiosity as its consequence, interference among rival explanations, and the endogenous menu) is about that segment.[11]

Every signal the agent encounters triggers two distinct processes in parallel. Under *within-prior Bayesian updating*, the posterior over θ for each active prior k updates by Bayes' rule; this is automatic and requires no allocation choice, and its effect is to refine the precision of the posterior within whichever mental models the agent has activated. Under *across-prior weight reassignment*, the same signal provides evidence about which prior better generated the observation, the agent uses it to reassess $\mathbf{w}$, and its effect is to determine the activation of each mental model. The evidence caps (6) are the discipline on the second process.

2.4 The cost of holding several mental models

A single act must serve whatever mixture of mental models the agent holds. The *hedging cost* of holding weights $\mathbf{w}$ is the value lost because the agent must choose one act rather than the act each mental model

Tonegawa, 2011), so a rival can lower the cap without ever being entertained.

[11] Given the weights, the agent's beliefs about θ are the mixture $\sum_k w_k \pi_k$, a single distribution, and with φ linear the value of information is the expected-utility value under that distribution. The two-model language is therefore not needed to describe what she believes; it is needed to describe what governs it. A Bayes factor exists only between identified components, so the cap cannot be stated for a single free distribution; the cognitive and hedging costs are functions of the weights, and the hedging cost is the expected value of perfect information about the component index, which a single prior has no counterpart to; and the updating (3) tracks which component each piece of evidence favors, which a single prior over θ cannot record. The paper's content lives in these three places, and none of them survives the collapse to a single prior.

recommends:

$$A(\mathbf{w}) \;:=\; \underbrace{\sum_{k=0}^{K} w_k\,\varphi\big(U_k\big)}_{\text{value of knowing the mental model}} \;-\; \underbrace{\max_{a}\sum_{k=0}^{K} w_k\,\varphi\big(\mathbb{E}_k u(a,\theta)\big)}_{\text{value of one act under model uncertainty}}, \tag{8}$$

where $U_k := \max_a \mathbb{E}_k u(a,\theta)$ is the model-contingent value. The hedging cost is nonnegative, zero when all weight rests on one mental model or when every model receiving weight induces the same act, and concave in $\mathbf{w}$ (Lemma OA.1 in the Online Appendix). It is the only force in the model that makes holding several mental models intrinsically uncomfortable, independently of any cognitive cost, and it is that force which keeps endogenous weights from collapsing into wishful thinking. The next subsection computes A in closed form.

2.5 A closed-form specification

Let the act be a scale of engagement $a \in \mathbb{R}$ with

$$u(a,\theta) \;=\; a(\theta-\kappa) - \tfrac{\lambda}{2}a^2, \qquad \lambda > 0, \tag{9}$$

where κ is the marginal cost of engagement and λ an adjustment cost. Costly signals are conditionally i.i.d. draws $y_i = \theta + \eta_i$, $\eta_i \sim \mathcal{N}(0, \nu_y^2)$, so mental model k's posterior precision after x and n signals is $P_k(n) = \tau_k + \nu_k^{-2} + n\nu_y^{-2}$ and, because today's belief is the average of the beliefs one might hold tomorrow, the variance of the step-(4) posterior mean, viewed from step (2), is

$$\psi_k(n) \;=\; \frac{1}{P_k(0)} - \frac{1}{P_k(n)}, \qquad \psi_k' > 0, \quad \psi_k'' < 0. \tag{10}$$

Under (9), the model-contingent value is available in closed form:

$$U_k(x,n) \;=\; \frac{(m_k-\kappa)^2 + \psi_k(n)}{2\lambda}, \qquad \frac{\partial U_k}{\partial n} \;=\; \frac{1}{2\lambda\,\nu_y^2 P_k(n)^2} \;>\; 0, \tag{11}$$

strictly increasing and strictly concave in n. I assume that the default remains the most precise model after the observation, $P_0(0) > P_k(0)$ for every $k \geq 1$, which holds whenever $\tau_0 > \tau_k$ and the loadings are equally strong, $\nu_k = \nu_0$. Then $P_k(n) < P_0(n)$ for every alternative, and signals are worth strictly more under any alternative than under the default at every n.

In the baseline $\varphi(U) = U$. Then (8) evaluates to

$$A(\mathbf{w}) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(m)}{2\lambda} \;=\; \frac{1}{2\lambda}\sum_{k=0}^{K} w_k\,(m_k - \bar{m})^2, \qquad \bar{m} := \sum_{k=0}^{K} w_k m_k, \tag{12}$$

the weighted variance of the model-implied posterior means, divided by the adjustment cost. It is zero when the agent is certain of one mental model and larger the more the mental models she holds disagree about θ. When a single alternative k is held, (12) reduces to

$$A \;=\; w_k(1-w_k)\,D_k(x), \qquad D_k(x) \;:=\; \frac{(m_k-m_0)^2}{2\lambda}, \tag{13}$$

which is maximal at $w_k = 1/2$. Call $D_k(x)$ the *disagreement* between alternative k and the default. It is not a parameter of the agent: m_k and m_0 are posterior means, each formed by reading the observation through its own attention mapping, so $D_k(x)$ measures how far apart the two mental models' recommended actions are *given the signal*, and the same agent faces a different D_k for different observations and for different

alternatives. Its $w(1-w)$ shape is the same one that the source function of Abdellaoui et al. (2011) and Baillon et al. (2018) attaches to perceived ambiguity, which is what makes interior weights, rather than a diffuse single prior, the object those indices are picking up. With several alternatives, (12) equals $\sum_{0\le j<k\le K} w_j w_k D_{jk}$ with $D_{jk} := (m_j - m_k)^2/2\lambda$: curiosity sums the disagreement between every pair of mental models held, including pairs of alternatives. The objective becomes

$$V(\mathbf{w}, n) \;=\; \sum_{k=0}^{K} w_k U_k(x,n) \;-\; A(\mathbf{w}) \;-\; C(\mathbf{w}) \;-\; cn, \tag{14}$$

and, when a single alternative k is held at weight w_k,

$$V(w_k, n) \;=\; (1-w_k)\,U_0(x,n) + w_k U_k(x,n) \;-\; w_k(1-w_k)D_k(x) \;-\; \tfrac{\gamma}{2}w_k^2 \;-\; cn. \tag{15}$$

Equation (15) makes the economic choice explicit. Weight on an alternative buys the value gap $U_k - U_0$, the extra payoff the alternative promises. It pays two costs: the hedging cost $w_k(1-w_k)D_k$ of acting on a compromise between mental models that recommend different actions, and the cognitive cost $(\gamma/2)w_k^2$ of holding a second mental model. Information enters in two ways: signals refine each mental model's estimate, which raises U_k through ψ_k, and signals that discriminate between the mental models shrink the model uncertainty the compromise insures against, both at a price of c per signal. The evidence cap then limits the choice from one side. The agent may place less weight on an alternative than its bound, and does so whenever the alternative promises too little to pay for these costs. She may never place more, so no promise, however large, buys weight the evidence does not support.

3 Threshold Effects in Alternative Activation and Curiosity Failures

This section characterizes threshold effects in alternative activation and the curiosity failures they produce. I first argue that even though the menu contains multiple mental models, shifts in mental models happen one alternative at a time. An alternative is activated when its value gain covers the costs of holding it, its weight is capped by the evidence constraint, and information is sought only once it is activated. Together these create threshold effects, below which a curiosity failure occurs. I then show what signals clear the threshold, give a formal definition of curiosity as the value of resolving the model uncertainty that activation creates, examine how ambiguity aversion changes activation and the motive for search, and characterize where the decision maker seeks information.

3.1 One alternative at a time

Before characterizing activation it is worth asking how the agent spreads attention across alternatives, because the answer simplifies everything that follows. For a given total attention off the default, the cost (4) is uniquely minimized by placing all of it on a single alternative, and any split pays an interference premium of $(\eta - \gamma)\sum_{j<k} w_j w_k$ (Lemma OA.2 in the Online Appendix). The agent's problem therefore has a built-in preference for entertaining one alternative at a time, the endogenous counterpart of a restriction imposed by assumption elsewhere in the literature (Schwartzstein and Sunderam, 2021; Gabaix, 2014). On the face of the simplex where a single alternative k is held, the objective is (15): the problem is a comparison between alternative k and the default, with the disagreement $D_k(x)$, the value gap $\Delta U_k := U_k(x,n) - U_0(x,n)$, and the evidence cap $\overline{W}_k(x)$ all computed from the observation. The agent's choice of *which* alternative to entertain is then the choice of the face on which (15) is highest, and the observation makes that choice for her through the same three objects: it decides which alternatives clear their evidence caps, how much each is worth, and how far each stands from the default. Mindset shifts are therefore one-alternative-at-a-time affairs, and the next subsections characterize them on a given face before returning to the interaction between alternatives.

3.2 Threshold effects and curiosity failures

The decision maker places weight on an alternative to the extent that acting on it promises, if it is right, a higher expected payoff than acting on the default, net of the costs of holding it, subject to the evidence constraint. Much like collateral constraints in models of poverty traps (Galor and Zeira, 1993; Banerjee and Newman, 1993), the evidence constraint generates threshold effects in alternative activation. This subsection derives them in three steps: the value condition and the evidence constraint, the complementarity between weights and information, and the threshold the two create, below which a curiosity failure occurs.

The value condition and the evidence constraint. Because the default is precise, its posterior mean is anchored to μ_0 and responds modestly to the data: a favorable observation tends to be absorbed into noise rather than shifting the value of the default categorization.

Activation of alternative k is governed by a value condition and the evidence constraint. The *value condition* is that the value gap ΔU_k clear a threshold T_k, so that the alternative is worth enough to pay for holding both models.[12] The *evidence constraint* is that the weight on k cannot exceed $\overline{W}_k(x)$, its evidence cap, which is near zero when x is consistent with the default. The value gap net of the threshold gives the desired weight, and $\overline{W}_k(x)$ caps how much of it is admissible. Which form the value condition takes depends on the curvature of (15) in w_k, namely $2D_k - \gamma$: the hedging cost is concave and destabilizes interior weights, while the cognitive cost is convex and stabilizes them, so T_k is stated regime by regime.

Proposition 1 (Activation)**.** *Fix n and an alternative k held on its own, and write $D_k = D_k(x)$ and $\overline{W}_k = \overline{W}_k(x)$. Activation of k requires the value condition $\Delta U_k > T_k$, and the resulting weight never exceeds $\overline{W}_k$. The threshold T_k and the form of the optimum depend on the curvature $2D_k - \gamma$.*

(a) (Graded regime, $\gamma > 2D_k$.) *$T_k = D_k$, V is strictly concave in w_k, and*

$$w_k^* \;=\; \mathrm{med}\Big\{\, 0,\; \frac{\Delta U_k - D_k}{\gamma - 2D_k},\; \overline{W}_k \,\Big\}.$$

Activation occurs if and only if $\Delta U_k > D_k$, and w_k^* is continuous and increasing in ΔU_k.

(b) (All-or-nothing regime, $\gamma < 2D_k$.) *V is strictly convex in w_k, so $w_k^* \in \{0, \overline{W}_k\}$, and $w_k^* = \overline{W}_k$ if and only if*

$$\Delta U_k \;>\; T_k \;=\; T_k(\overline{W}_k) \;:=\; D_k\,(1 - \overline{W}_k) + \tfrac{\gamma}{2}\overline{W}_k \;<\; D_k.$$

The threshold is strictly below D_k: when the alternative and the default disagree enough, a decisive shift can pay even though no partial shift would. The threshold T_k is strictly decreasing in $\overline{W}_k$, and at the threshold w_k^* jumps discontinuously from 0 to $\overline{W}_k$.

(c) (Knife edge, $\gamma = 2D_k$.) *$T_k = D_k$, V is linear in w_k, and $w_k^* = \overline{W}_k$ if $\Delta U_k > D_k$, $w_k^* = 0$ otherwise.*

Both $\overline{W}_k(x)$ and $D_k(x)$ are functions of the observation, so the observation determines which alternatives can be activated and, for each, whether the weight on it moves continuously or in switches.

Figure 1 illustrates Proposition 1. Panel A shows why the all-or-nothing regime produces a jump: when the hedging cost outweighs the cognitive cost, the gain from activation is convex in the weight, so partial activation is never optimal. Panel B shows the two resulting paths for the optimal weight. The evidence cap is the first part of the proposition: an alternative whose bound (6) the observation leaves near zero can carry only negligible weight whatever its value, and because each bound depends on how the observation is read

[12] "Worth" here means decision value, not optimism. Under (11), U_k rises with how decisive the action model k recommends is, $(m_k - \kappa)^2$, and with what remains to be learned under it, $\psi_k(n)$. An alternative that reads the opportunity as harmful, and so calls for decisively opting out, can therefore carry a larger value gap than a lukewarm default. What the agent never activates is an alternative that would change her action too little, or teach her too little, to pay for holding it.

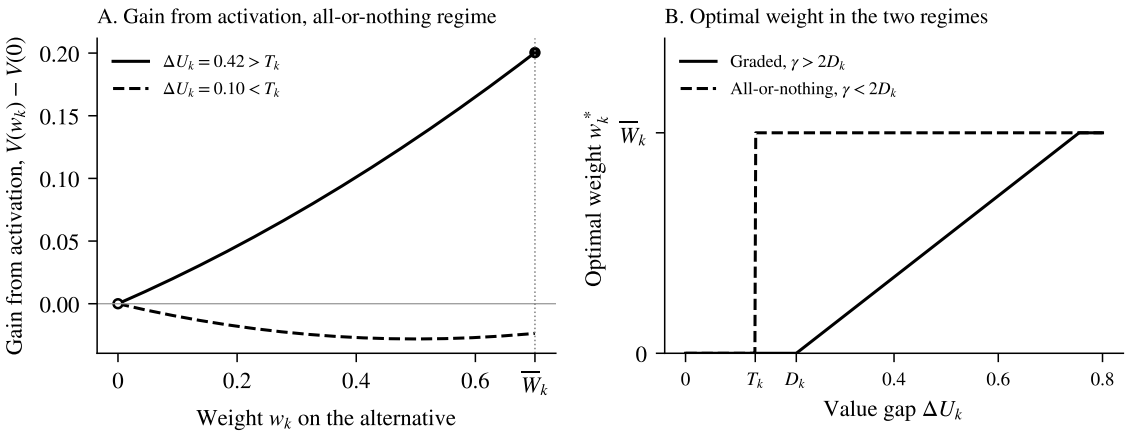


FIGURE 1. THRESHOLD EFFECTS IN ALTERNATIVE ACTIVATION

Notes: Both panels illustrate Proposition 1 with $D_k = 0.2125$ and evidence cap $\overline{W}_k = 0.70$. Panel A plots the gain from activation, $V(w_k) - V(0)$ from (15), in the all-or-nothing regime ($\gamma = 0.20 < 2D_k$). The gain is convex in the weight, so the optimum is a corner: for a value gap above the threshold $T_k(\overline{W}_k) = 0.134$ (solid) the decision maker moves straight to the evidence cap, and for a value gap below it (dashed) any partial activation is worse than none. Panel B plots the optimal weight against the value gap in the two regimes. In the graded regime ($\gamma = 1.20 > 2D_k$) the weight rises continuously once $\Delta U_k > D_k$ and stops at the evidence cap; in the all-or-nothing regime it jumps from zero to the evidence cap at $T_k(\overline{W}_k)$.

and not only on what it contains, contrast decides which alternative is activated. The value condition is the second, and its threshold depends on a regime the observation also decides. A signal that alternative k reads as calling for roughly what the default calls for leaves m_k near m_0, so D_k is small and the agent is in the graded regime; a signal it reads as calling for something different pushes m_k away from m_0 and she is in the all-or-nothing regime. The same agent can therefore adjust continuously after one observation and switch after the next. Note also that activation requires more than a value advantage: in the graded regime the condition is $\Delta U_k > D_k$, not $\Delta U_k > 0$, so alternatives that disagree sharply with the default are *harder* to activate partially.

The two regimes are the counterparts of the gradual and abrupt transitions neuroscientists observe when the neurons encoding an animal's situation meet discrepant evidence: *rate remapping*, in which cells keep their firing fields and change only their rates (Leutgeb et al., 2005a), and *global remapping*, the all-at-once switch to an independent code (Wills et al., 2005; Leutgeb et al., 2005b; Colgin et al., 2008) that Sanders et al. (2020) interpret as Bayesian inference about which situation the animal is in. The model generates the dichotomy rather than assuming it, though nothing here establishes that its classifying variable is what those experiments manipulate.

The dichotomy does not depend on the quadratic payoff, the Gaussian structure, or a linear φ: the objective is a difference of convex functions, and which term has the greater curvature decides whether the weight moves continuously or sits at a corner (Remark OA.1 in the Online Appendix). The constraint is what separates the model from one of motivated beliefs, and the separation can be stated exactly: without the cap, the optimal weight is independent of the likelihood ratio, so an alternative the observation contradicts arbitrarily strongly is adopted in full whenever it is worth enough (Proposition OA.2 in the Online Appendix).

Without the cap the agent chooses her categorization for what it delivers, as beliefs are chosen in optimal expectations (Brunnermeier and Parker, 2005; Bénabou and Tirole, 2016); the hedging and cognitive costs temper how far she goes but not what she goes toward.[13] The cap makes evidence, not value, the gatekeeper, and it is the only feature of the model that does so.

[13]The correspondence is structural rather than motivational: there the agent chooses any distribution and gains anticipatory utility from optimism, here she chooses among categories she already holds and gains nothing from holding them. The menu restricts her to motivated categorization rather than motivated belief; the cap restricts her further, to categories the evidence supports.

A consequence is that isolated flukes do not produce small durable weights: in the all-or-nothing regime an observation that falls short of the threshold leaves the weight at zero exactly, and in the graded regime it produces a weight too small to prompt the information that would let it grow (Corollary OA.1 in the Online Appendix).

Complementarity between weights and information. Activation is the first half of the joint decision. The second is information acquisition, and the two are linked by a complementarity between w_k and n.

Proposition 2 (Complementarity)**.** *The effect of an additional signal on the return to attention to alternative k, the cross-partial of* (5)*, is*

$$\frac{\partial^2 V}{\partial w_k \, \partial n} \;=\; \varphi'(U_k)\frac{\partial U_k}{\partial n} - \varphi'(U_0)\frac{\partial U_0}{\partial n},$$

which is strictly positive if and only if $\big(P_k(n)/P_0(n)\big)^2 < \varphi'(U_k)/\varphi'(U_0)$*. Each of the following is sufficient:* (i) φ *linear;* (ii) $U_k \le U_0$*;* (iii) *signals that the default does not read,* $\partial U_0/\partial n = 0$*. When the inequality holds,* (w_k, n) *are complements:* w_k^* *is nondecreasing in n and* n^* *is nondecreasing in* w_k*, and the first-order condition in n is* $\sum_{j=0}^{K} w_j \, \varphi'(U_j) \, \partial_n U_j = c$*.*

Complementarity is a condition rather than an identity: signals are worth more under the diffuse model, which is what $P_k < P_0$ says, and this must outweigh the curvature of φ. The sharper claim that $n^* = 0$ whenever $\mathbf{w}^* = 0$ needs one further assumption.

Assumption 1 (Alternative-specific signals)**.** The costly signals load on features the default mental model does not encode: $\partial U_0/\partial n = 0$.

Under Assumption 1 complementarity is unconditional and the marginal benefit of a signal at $\mathbf{w} = 0$ is exactly zero, so $n^* = 0$: there is no point acquiring detail about a model that has not been opened. Information seeking is thus absent under a single mental model, however diffuse its prior, and grows with the attention the alternative receives. Assumption 1 is used only for this corner result and its consequences. Without it the comparative static $\partial n^*/\partial w_k \ge 0$ survives but the corner result does not. The information side is then also available in closed form.

Corollary 1 (Closed-form information acquisition)**.** *Under Assumption 1 with* φ *linear, and treating n as continuous, the optimal number of signals when alternative k is held on its own at weight* w_k *is*

$$n^*(w_k) \;=\; \nu_y^2 \Big[\Big(\frac{w_k}{2\lambda \nu_y^2 c} \Big)^{1/2} - P_k(0) \Big]^+,$$

which is zero for $w_k \le 2\lambda \nu_y^2 c \, P_k(0)^2$ *and thereafter increasing and concave in* w_k *and decreasing in c. Together with Proposition 1, the optimum* (w_k^*, n^*) *is the fixed point of two explicit best responses.*

Threshold effects and curiosity failures. Corollary 1 has a threshold: no signal is bought until the weight on the alternative exceeds $2\lambda \nu_y^2 c \, P_k(0)^2$, a level that rises with the cost of signals and with the alternative's own precision. Consider an alternative that is more promising than the default and right. If evidence kept arriving, its evidence cap would rise steadily and the weight would follow. Two features break this convergence. First, within a stage the initial signal fixes the evidence cap. Second, weights and information are complements (Proposition 2), so information is sought only once the weight is high enough. Together they create a threshold. When the observation supports the alternative only weakly, the cap holds the weight below the level at which information pays: the alternative is held but not investigated, stays imprecise, and its cap never rises. This is a *curiosity failure*: the decision maker never seeks the information that could change her mind. When the observation lifts the weight above that level, a self-reinforcing cycle begins: the weight

justifies information, the information raises the cap through the update (3), and at the next stage the higher cap justifies more weight. Small differences in the initial signal can thus produce sharply different outcomes. Parents may ignore a school-based agricultural education program if it fails to distinguish itself from school gardens that use junior high students as farm labor, while a light-touch information intervention can prompt them to seek information on their own (Lee, 2026). Above the threshold, the path depends on the regime of Proposition 1: when the alternative recommends much the same action as the default, the weight tracks the evidence continuously and information is bought as it rises; when it recommends a different action, the weight switches to its bound and information follows.

3.3 What signals clear the threshold

Given the central role of the initial signal, which signals clear the threshold? The answer is not simply those that supply more observable features that fit the alternative. Unlike in models with one-dimensional signals, how evidence is framed is central here, for two reasons: the evidence constraint rests on relative likelihoods, and a signal is a vector of observable features that each mental model reads through its own attention. Because (1) is a linear Gaussian model, each L_k is available in closed form and each Bayes factor decomposes into three interpretable channels, which turns the evidence constraint into a design instrument. Under (1), $x \sim \mathcal{N}(\mu_k b_k,\ \Sigma + \sigma_k^2 b_k b_k')$ under mental model k, and $L_k(x)$ is that density evaluated at x (Lemma OA.4 in the Online Appendix).

Proposition 3 (Three-channel decomposition of the evidence constraint)**.** *Define, for each mental model* $k = 0, 1, \ldots, K$*,*

$$\underbrace{R_k := x'\Sigma^{-1}x - s_k^2/\nu_k^2}_{\textit{off-attention residual}}, \qquad \underbrace{q_k := \frac{(s_k-\mu_k)^2}{\sigma_k^2+\nu_k^2}}_{\textit{standardized surprise}}, \qquad \underbrace{c_k := \log\left(1+\frac{\sigma_k^2}{\nu_k^2}\right)}_{\textit{Occam factor}}.$$

Then $R_k \geq 0$ *and, for each alternative* k*,*

$$\log \Lambda_k(x) \;=\; \tfrac{1}{2}\Big[\underbrace{(R_0 - R_k)}_{\textit{encoding}} + \underbrace{(q_0 - q_k)}_{\textit{surprise}} + \underbrace{(c_0 - c_k)}_{\textit{Occam}}\Big]. \tag{16}$$

Consequently: (i) *the encoding channel vanishes identically if and only if* $b_0 \propto b_k$*;* (ii) *the Occam channel is strictly negative whenever* $\sigma_k^2/\nu_k^2 > \sigma_0^2/\nu_0^2$*, in particular whenever the alternative has the default's loading,* $b_k = b_0$*, and the more diffuse prior; and* (iii) $\overline{W}_k(x)$ *is increasing in each of the three channel differences of alternative* k*, decreasing in each channel difference of every other alternative, and equals the Bayesian posterior model probability under a uniform prior over mental models.*

The decomposition identifies three margins on which a signal can raise the evidence cap of an alternative. The first is *surprise*, the channel $q_0 - q_k$: the alternative is less surprised by the part of the signal it reads than the default is by the part it reads, as when it better predicts an observation that is extreme on a feature both mental models read. The second is *encoding*, the channel $R_0 - R_k$: the signal carries features that the alternative reads but the default ignores. The third is *specificity*, the Occam channel $c_0 - c_k$. A marginal likelihood penalizes a model for spreading predictive mass, and c_k is the Occam factor of Bayesian model comparison (MacKay, 1992; Jefferys and Berger, 1992; Kass and Raftery, 1995; Ba, 2026). Every alternative therefore begins at a handicap relative to the more precise default, and the handicap shrinks as the alternative's prior approaches the default's in precision. Since c_k is increasing in σ_k^2, an optimistic but vague alternative ("the program might really help") is *harder* to justify than a specific one ("pass rates rise by about eight points"), even when the vague version has the higher mean. Wishfulness does not buy likelihood; precision does. Precision has two limits. A narrower alternative whose predictions miss loses more likelihood than a

vague one. And precision lowers value: a more precise prior leaves less to learn, so $\psi_k(n)$ and with it the value gap ΔU_k fall. Narrowing an alternative therefore raises its evidence cap but can weaken the value condition, and it helps most when the cap binds.

Two implications follow. First, exploration can be stimulated while holding constant the observable features the default reads, by presenting features that only an alternative reads (encoding) or by stating the alternative as a specific, checkable claim consistent with what is observed (specificity). Second, a signal can fail to change behavior not for lack of observable features but for lack of any mental model that reads or explains them: a feature no mental model on the menu reads leaves every channel, and hence every evidence cap, unchanged. For information interventions, the first implication means that an intervention can activate an alternative without adding to the evidence the default already reads: showing parents what students can do on the school farm, which the default ignores, or stating a specific, checkable claim about pass rates can do more than further yield data. The second means that evidence alone may not be enough: an intervention may first need to supply a mental model that reads it, for instance by explaining why a technique works.

3.4 A formal definition of curiosity

Activation creates model uncertainty, and curiosity is the drive to resolve it. Curiosity is not left as a description. The hedging cost already defined is the value of learning which mental model is right at the weights the agent holds, so it can serve as the definition.

Definition 1 (Curiosity)**.** At weights $\mathbf{w}$, *curiosity* is the value of resolving which mental model is right,

$$A(\mathbf{w}) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(m)}{2\lambda},$$

the component of the value of information that comes from discriminating between the models rather than from refining either.

Curiosity is not a separate choice. It is a function of the weights, and the agent's curiosity is its value at the weights she chooses. Because the weights are chosen jointly with information, curiosity is determined with them, and it shows up in behavior through the information she acquires: how much and along which features.

Three properties follow. Curiosity is zero at $\mathbf{w} = \mathbf{0}$, so an agent under the default alone is not curious however diffuse that model is; on a face it equals $w_k(1-w_k)D_k$ and so rises with the weight placed on the alternative over the relevant range; and it values only signals that discriminate between the mental models, not signals that refine both. Curiosity thus coincides with the hedging cost of Section 2.4: the cost of model uncertainty and the value of resolving it are one object.

Definition 1 shares the structure of the information-gap account but not, in the baseline, its motive. In that account curiosity is the aversive state produced by awareness of a gap between what one knows and what one wants to know (Loewenstein, 1994), and it is modeled through preferences over beliefs, with attention to a question taken as given (Golman and Loewenstein, 2018). Three correspondences are close. A gap requires a question, and here the question does not exist until an alternative is activated, so the definition supplies the step that account assumes. Intensity is inverted-U in knowledge there, and $w_k(1-w_k)D_k$ is inverted-U in the weight here, greatest when the agent is least sure which model applies and vanishing at either extreme. And curiosity is directed at a particular question there, while here it directs the agent toward the observations that would settle which mental model is right. The difference is the motive. With φ linear, A is instrumental: it values only information that could change what the agent does, so the appetite for an answer that bears on nothing lies outside it; that restriction follows from the design problem, since an intervention can put an alternative in front of someone but cannot install an appetite for information in general. The non-instrumental motive enters with ambiguity aversion, which makes the agent value resolving which mental model is right even when the answer would not change her action. Nor is A the value of information as such: the value of

refining either mental model is available under a single model and is not curiosity in the sense used here.[14]

The next result completes the case for Definition 1. Suppose the act is evaluated at the weights updated by (3). Let $\bar{m}_n := \sum_{k=0}^{K} w_{k,n}\, m_{k,n}$ be the mixture posterior mean after the signals, and let $A_n := \mathrm{Var}_{\mathbf{w}_n}(m_{\cdot,n})/(2\lambda)$ be the hedging cost the agent faces at step (4), the analogue of (12) at the updated weights and means.

Proposition 4 (The value of information with updating)**.** *Under* (9) *with* φ *linear and the update* (3)*, the ex-ante value to the agent of the n signals, evaluated under the mixture* $\mathbf{w}$*, is*

$$\mathrm{VOI}(\mathbf{w},n) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(\bar{m}_n)}{2\lambda} \;=\; \underbrace{\frac{\sum_{k=0}^{K} w_k\, \psi_k(n)}{2\lambda}}_{\textit{within-model refinement}} \;+\; \underbrace{A(\mathbf{w}) - \mathbb{E}_{\mathbf{w}}[A_n]}_{\textit{resolution of model uncertainty}} \quad . \tag{17}$$

(a) $\mathrm{VOI}(\mathbf{w},n) \geq 0$ *for every* $(\mathbf{w},n)$*: the agent is never averse to information.*

(b) The resolution component vanishes at every vertex of the simplex and converges to the hedging cost, $A(\mathbf{w}) - \mathbb{E}_{\mathbf{w}}[A_n] \to A(\mathbf{w})$ *as* $n \to \infty$*: in the limit, information recovers exactly the cost of holding several mental models at once.*

(c) Under Assumption 1, $\mathrm{VOI}(0,n) = 0 < \mathrm{VOI}(\mathbf{w},n)$ *for every* $\mathbf{w} \neq 0$ *and* $n \geq 1$*: curiosity is absent under the default alone and present once an alternative is activated.*

(d) Model uncertainty is resolved in one of two ways. Take a single alternative k held against the default. If both mental models read the signals as informative about θ*, then* $m_{k,n} - m_{0,n} \to 0$ *while* $w_{k,n}$ *converges almost surely to a limit strictly between* 0 *and* 1*: the models come to recommend the same action, so the hedging cost disappears, but the question of which model is right is never settled. Under Assumption 1,* $w_{k,n} \to 1$ *almost surely if the signals are generated as the alternative says and* $w_{k,n} \to 0$ *if they are generated as the default says: the question is settled and the agent keeps one mental model.*

Part (b) makes precise what the introduction calls resolving the model uncertainty that activation creates: the hedging cost is by construction the value of learning which mental model is right, and enough signals deliver that value in full. Part (d) separates two things "resolving model uncertainty" can mean: the models coming to agree, or one of them winning. Part (a) is a statement about a linear φ; under a concave φ the ex-ante value of information is not signed in general (Li, 2020), and the results on the weights chosen at step (2) do not depend on its sign.

3.5 Ambiguity aversion

The closed forms set φ linear. A strictly concave φ adds ambiguity aversion, which changes both whether an alternative is activated and what the decision maker is curious about. Both comparisons below are made at the time of choice, so neither relies on the dynamic consistency of the smooth ambiguity criterion (Klibanoff et al., 2009).

Proposition 5 (Ambiguity aversion and activation)**.** *Let* φ *be strictly concave, let* $\bar{U} := (1-w_k)U_0 + w_k U_k$*, and let* $\rho := -\varphi''(\bar{U})/\varphi'(\bar{U})$*. To second order in* ΔU_k*,*

$$(1-w_k)\varphi(U_0) + w_k\varphi(U_k) \;=\; \varphi(\bar{U}) - \tfrac{\rho}{2}\,\varphi'(\bar{U})\, w_k(1-w_k)(\Delta U_k)^2,$$

so holding alternative k carries an ambiguity premium $\frac{\rho}{2}\varphi'(\bar{U})\, w_k(1-w_k)(\Delta U_k)^2$ *beyond the hedging cost.*

[14] Given the weights, the value of information here is an expected-utility object: the value of refining the mixture belief plus the expected value of perfect information about which model is right. What is not an expected-utility object is activation, the choice of the weights, and it is activation that makes curiosity switch on, scale with the weight, and be absent under the default. A Bayesian who holds two models is always curious; the agent here is curious only once she has chosen to hold a second one.

Proposition 1 holds with D_k replaced by $\tilde{D}_k := D_k + \frac{\rho}{2}(\Delta U_k)^2$: the activation thresholds rise, and the all-or-nothing regime obtains on a set of parameters that grows with ρ.

Ambiguity aversion thus makes alternatives harder to activate, and abrupt shifts more likely, the larger the stakes $(\Delta U_k)^2$. The premium also adds a non-instrumental motive for curiosity. It is positive even when the two mental models recommend the same action, so that $D_k = 0$ and resolving which is right has no instrumental value. An ambiguity-averse decision maker therefore values resolving the question for its own sake, in proportion to the stakes $(\Delta U_k)^2$. This is the non-instrumental motive of the information-gap account (Loewenstein, 1994; Golman and Loewenstein, 2018), confined here to questions that bear on payoffs.

3.6 Endogenous exposure: where the decision maker seeks information

So far each signal updates the default and the alternatives alike. From here on, each signal also has a direction d, $y = d'\tilde{x} + \eta$ with $\eta \sim \mathcal{N}(0, \nu_y^2)$, so that the agent chooses what information to look at in addition to how much to acquire. Directions may include those the default reads. She chooses the direction first and then how many observations to make. A school visit that looks at crop yields is one d; one that looks at what students can do is another. With a single alternative k held at weight $w = w_k$, write $\delta := b_k m_k - b_0 m_0$ for the difference in the two models' predictions, $\Omega_0 := \Sigma + b_0 b_0'/P_0(0)$ for the default's predictive covariance, and

$$v(w) \;:=\; \underbrace{(1-w)\,\frac{b_0}{P_0(0)} + w\,\frac{b_k}{P_k(0)}}_{\text{refinement}} \;+\; \underbrace{w(1-w)\,(m_k - m_0)\,\delta}_{\text{resolution}}.$$

Proposition 6 (What the agent looks at)**.** *For an observation that is weak relative to the agent's prior uncertainty, with predictive variances common to the two models, the value of an observation along d is proportional to $(d'v(w))^2/(d'\Omega_0 d + \nu_y^2)$. Among directions normalized by $d'\Omega_0 d = 1$, the agent therefore looks along*

$$d^o(w) \;\propto\; \Omega_0^{-1}\, v(w).$$

(i) *At $w = 0$ she looks through the default's lens, $\Omega_0^{-1} b_0$, and at $w = 1$ through the alternative's, $\Omega_0^{-1} b_k$.* (ii) *In between, the pull toward the direction of disagreement $d^* \propto \Omega_0^{-1}\delta$, the direction that most reduces the agent's model uncertainty, $w(1-w)$, is proportional to $w(1-w)(m_k - m_0)$: it is strongest when model uncertainty is greatest and when the answer would change what she does.* (iii) *If the two models recommend the same action, $m_k = m_0$, the pull vanishes, however much their predictions differ.*

The proposition turns the choice of what to look at into a tradeoff between learning more within the mental models she holds and learning which of them is right, and the weight on the alternative sets the terms. When the alternative carries little weight, she looks where the default looks, and little, because the precise default has little left to learn and there is little model uncertainty to resolve: seeing what one expects to see is the optimal use of attention for someone with little model uncertainty. The value of an observation rises with the weight on the vaguer alternative, through both terms, which is the complementarity of weights and information in another form. As model uncertainty grows, she is pulled toward the features on which the mental models disagree, and once the alternative dominates, she looks through its lens to learn how to act on it. A farmer who credits the rainfall watches the weather; one who suspects a new practice compares plots; one who has adopted it studies how to apply it. The resolution term is curiosity at work. It vanishes under the default alone and when both mental models recommend the same action, so curiosity targets the disagreements that matter for what she does, and it mirrors the directedness of curiosity in the information-gap account (Loewenstein, 1994): the gap is the predictive disagreement between the mental models held. Information gathered through the default's lens at low weight is routine refinement, not curiosity; what is absent under the default alone is information about which mental model is right.

Ambiguity aversion and the direction of search. The ambiguity premium of Proposition 5 also changes where the decision maker looks, since a signal changes the premium through the weight and the value gap it leaves behind.

Proposition 7 (Ambiguity aversion tilts search toward discriminating signals)**.** *Under the second-order expansion, the value of n signals to an ambiguity-averse agent equals its value under linear φ plus*

$$\tfrac{\rho}{2}\,\varphi'(\bar{U})\Big[(\Delta U_k)^2\, w_k(1-w_k)\;-\;\mathbb{E}\big[(\Delta U_{k,n})^2\, w_{k,n}(1-w_{k,n})\big]\Big].$$

(i) *A signal that moves only the weight, one that discriminates between the models without refining either, makes this term nonnegative, because $w_{k,n}$ is a martingale and $w(1-w)$ is concave.* (ii) *A signal that moves only the posterior means, one that refines both models without discriminating between them, makes it nonpositive, because $\Delta U_{k,n}$ is a mean-preserving spread of ΔU_k. Ambiguity aversion therefore raises the value of discriminating signals and lowers the value of refining ones, and the tilt is increasing in ρ.*

An ambiguity-averse decision maker thus seeks more information along the direction of disagreement of Proposition 6 and less about outcome data both mental models can absorb than a linear agent with the same beliefs would. The smooth ambiguity aggregator, which plays no role in the closed forms, is what sets the *direction* of search: the more the decision maker dislikes model uncertainty, the more her information seeking is aimed at ending it rather than at refining the mental model she favors. Because the premium grows with $(\Delta U_k)^2$, both of its effects are largest for questions with high stakes: such questions are harder to open, since the premium raises the activation threshold, but once open they are pursued more actively. Where stakes are high, interventions that activate an alternative therefore gain relative to those that lower the cost of information. For example, when farmers suspect that fertilizer bought on credit may be fake (Bold et al., 2017; Michelson et al., 2021), the model predicts that side-by-side plots grown from the same bag or nutrient tests at the point of sale do more than cheaper credit, and that small trial packs, by lowering the stakes, make the question easier to open. The comparison across stakes requires that absolute ambiguity aversion ρ not fall in proportion to them; if relative ambiguity aversion is constant instead, the premium and the value gap scale together.

4 Behavioral Effects

Standard decision problems take the decision maker's representation of the situation as given. Associative-memory models, building on the psychology of memory (Howard and Kahana, 2002; Kahana, 2012), instead make that representation an outcome of recall: what comes to mind depends on its similarity to the situation, the cues present, and competition from other memories, and beliefs and choices follow from what is retrieved. One group of studies shows how recall shapes beliefs: people judge probabilities by the scenarios that come to mind (Gennaioli and Shleifer, 2010; Bordalo et al., 2023), imagine the future by drawing on memory (Bordalo et al., 2025), and react to news through cue-driven recall (Enke et al., 2024). Another shows how recall shapes choice: recalled experiences set the norm against which an option is judged (Bordalo et al., 2020), and a problem is recognized by retrieving the most similar category of past problems (Bordalo et al., 2026). Framing and context thus matter because they change what comes to mind, and so which problem the decision maker believes she is solving.

The framework here shares that premise. Associative memory supplies the default and the menu, and the attention mapping of the mental model in use determines how data are read. But because the decision maker can seek information and reason beyond what associative memory supplies, the model revises two hallmark behavioral effects of associative-memory models, context effects and interference effects. This section illustrates how both are revised. Appendix A sets out the neuroscientific regularities the model reproduces and what distinguishes it from its closest alternatives.

4.1 Context effects

In psychology, a context effect is the finding that cues present when an experience is stored and when it is recalled shape what is remembered, even when the cues are irrelevant to the task (Tulving and Thomson, 1973; Godden and Baddeley, 1975; Smith and Vela, 2001); the temporal context model formalizes how context drives retrieval (Howard and Kahana, 2002; Kahana, 2012). Economic associative-memory models build on this finding: a payoff-irrelevant cue changes which experiences or representations come to mind, and so changes beliefs and choices (Gennaioli and Shleifer, 2010; Bordalo et al., 2020; Enke et al., 2024; Bordalo et al., 2026). For example, a consumer sees a jar of jam in plain packaging as a routine purchase and focuses on its price, but the same jar in fancy packaging becomes a treat and draws her attention to its quality.

The encoding channel of Proposition 3 determines when context matters here. Call a component z of the observation *payoff-irrelevant under the default* if the default does not read it, $b_0'\Sigma^{-1}z = 0$.

Corollary 2 (Context effects)**.** *Let x and $x+z$ be two observations, where z is payoff-irrelevant under the default.* (i) *If no mental model on the menu reads z, that is, $b_k'\Sigma^{-1}z = 0$ for every k, then $\Lambda_k(x+z) = \Lambda_k(x)$ for every k, so the evidence constraints, the weights, the information acquired, and the act are all unchanged.* (ii) *Otherwise z moves Λ_k only for the alternatives k that read it, through their statistics s_k, and moves the bound of any other alternative only through the denominator of* (6)*.*

The proof is immediate from (16): $x'\Sigma^{-1}x$ cancels from $R_0 - R_k = s_k^2/v_k^2 - s_0^2/v_0^2$, so every term of $\log\Lambda_k$ depends on x only through the statistics $s_j = b_j'\Sigma^{-1}x/(b_j'\Sigma^{-1}b_j)$, and a component no model reads leaves each of them unchanged. Part (ii) is the formal content of the encoding margin. Because the default does not read z, its statistic s_0, and with it the default's surprise q_0, is unchanged, and the Occam channel does not depend on the observation at all, so z can act only through the statistic s_k of an alternative that reads it. In associative-memory models, adding a payoff-irrelevant cue changes which representation of the situation comes to mind and so changes behavior (Bordalo et al., 2026). Here a cue sways behavior only if it is payoff-relevant under some mental model on the menu and lifts that model over the activation threshold of Proposition 1; if it does, the curiosity this triggers tests the shift, as the next result shows.

Corollary 3 (Exposure and the path of the shift)**.** *Let observations along the direction she chooses be generated by a density g, and let p_k and p_0 be the predictive densities of alternative k and the default along it. Each observation changes $\log\Lambda_k$ in expectation by $\mathrm{KL}(g\,\|\,p_0) - \mathrm{KL}(g\,\|\,p_k)$: the shift continues on the agent's own exposure if the alternative fits the observations better than the default, and reverses otherwise. If $g = p_k$, the change is $\mathrm{KL}(p_k\,\|\,p_0)$, positive whenever the two predictive distributions differ; if $g = p_0$, it is $-\mathrm{KL}(p_0\,\|\,p_k)$. At $w_k = 0$ no information about the alternative is sought and neither occurs.*

This delivers persistence from choices rather than from a state variable: a shift that crosses the threshold carries itself forward when the alternative fits the observations better than the default and corrects itself otherwise, because the agent now looks where the answer lies, while below the threshold her exposure stays governed by the default and nothing she sees moves the bound.

Remark 1 (Misdirected search)**.** *When one of the two mental models is true, reversal of a mistaken shift in Corollary 3 does not require the agent to look along d^*. If the default is true, then for any direction d an observation changes the log Bayes factor of the alternative, in expectation, by $-\mathrm{KL}(p_0\,\|\,p_k)$ along d, which is nonpositive and strictly negative whenever the two predictive distributions differ; and by Lemma OA.3(i) the odds of the alternative against the default cannot exceed its updated Bayes factor. The argument holds for any rule that selects d, optimal or not and possibly history-dependent, so it covers departures from the optimal direction of Proposition 6, such as a tendency to seek confirming evidence, that is, to favor directions in which the alternative predicts a confirming outcome. Such a distortion therefore slows correction, and stalls it only along directions on which the models make identical predictions, but cannot reverse its direction. The robustness requires that one of the two mental models be true. If neither is, the sign of $\mathrm{KL}(g\,\|\,p_0) -$*

$\mathrm{KL}(g \,\|\, p_k)$ *can depend on d, and a distortion of where the agent looks can select directions along which the alternative fits better, even though the default fits better along others, and so sustain the shift. Entrenching a shift therefore requires evidence that is biased toward it, or a menu on which no mental model is true combined with misdirected search.*

Corollary 2 separates two questions that associative-memory models answer together. Whether context moves behavior depends on the menu of mental models: some model must read the cue. Whether the movement lasts depends on the information environment and the menu. By Corollary 3 and Remark 1, a context effect that persists points to evidence that is scarce, too costly to seek given the weight the alternative carries (Corollary 1), or biased toward the mistaken model, or to a menu on which no mental model is true. Because the decision maker has an endogenous drive to seek this evidence herself, policy can make it accessible instead of delivering it: for farmers who distrust fertilizer that looks clumped but meets nutrient standards (Michelson et al., 2021), the remedy is quality testing they can obtain on their own.

4.2 Interference effects

In psychology, interference is the finding that similar memories compete at retrieval, so that recalling one item makes related items harder to recall (Underwood, 1957; Anderson and Neely, 1996; Anderson, 2003; Kahana, 2012). In economic associative-memory models, different experiences compete for retrieval, so recalling one inhibits recall of others and a prime can crowd out better information (Bordalo et al., 2023, 2025). For example, in Amazon's early days, people who thought of bookstores saw it as a firm in a mature sector: these frequent experiences crowded out the fewer but more relevant experiences of disruptive technologies, which would have suggested a future giant (Bordalo et al., 2025). In these models, the representation with the greatest cue similarity is the one that comes to mind.

By contrast, in this framework the menu delivers a result that no comparison of one alternative with the default can. When each of multiple alternatives is worth more than the default, none may be activated, because the same evidence must justify each of them and justifies none enough: the evidence cap of each alternative is deflated by every other alternative that looks plausible, the agent's attention is rivalrous across them, and the superadditive cost of holding several priors at once raises the price of search. Two forces make a menu of alternatives harder to act on than a single one: the cost function penalizes spreading attention (Lemma OA.2), and the evidence caps crowd one another. These are interference effects, but not the kind associative-memory models study, where the representation with the greatest cue similarity comes to mind. Here every alternative remains available, and rivals interfere by dividing the support the evidence provides. What interferes is thus a feature of the information environment, such as a cue that lends plausibility to a rival or evidence spread across several alternatives rather than concentrated on one, acting on a given menu, not a failure of retrieval.

Proposition 8 (Fragmentation)**.** *Fix x and let $\overline{W}_k$ be given by* (6)*.*

(i) (Crowding.) *For each $k \geq 1$, $\overline{W}_k$ is strictly decreasing in L_j for every $j \neq k$ with $j \geq 1$. In particular $\overline{W}_k < L_k/(L_0 + L_k)$, the bound that would obtain were k the only alternative on the menu.*

(ii) (Breadth dilutes.) *Let a treatment scale the likelihoods of a set $S \subseteq \{1, \ldots, K\}$ by a common factor $\chi > 1$. For $k \in S$, the resulting bound is strictly smaller when $|S| \geq 2$ than when $S = \{k\}$.*

(iii) (Joint failure.) *Consequently, in the all-or-nothing regime, a feature vector that raises several L_k moderately can leave every $\overline{W}_k$ below the level alternative k needs in Proposition 1, in which case $w_k^* = 0$ for all $k \geq 1$ and, under Assumption 1, $n^* = 0$. In the graded regime crowding caps each weight at $\overline{W}_k$, and no information is acquired once every $\overline{W}_k$ falls below the level in Corollary 1.*

Corollary 4 (Crowding by unpromising rivals)**.** *Fix x and let alternative j satisfy $\Delta U_j \leq \min\{D_j, \gamma/2\}$, so that by Proposition 1 it is never activated, whatever its bound. Without the evidence constraint, adding j to the menu leaves $(\mathbf{w}^*, n^*)$ unchanged. With it, adding j multiplies every other bound by $(1 + \sum_{i \neq j} \Lambda_i)/(1 +$*

$\sum_i \Lambda_i) < 1$, *so it can prevent the activation of an alternative k that would otherwise be activated and, under Assumption 1, set* $n^* = 0$.

Without the constraint, the weight on an alternative depends on its value gap and not on any likelihood, and ΔU_j lies below the threshold of both regimes, so j receives no weight and the problem for the other alternatives is unchanged. With the constraint, j still receives no weight, but it enters the denominator of (6) through Λ_j alone, whatever its value. The agent can ignore an unpromising explanation but not the evidence for it: a plausible story she would never entertain still crowds out one she would. Under optimal expectations such an explanation could simply be dropped from the menu; here it cannot, and adding an alternative that is never chosen can change the choice. The same holds for cues: a contextual cue that such an alternative reads but the promising one does not raises the former's likelihood and so, through the denominator of (6), lowers the promising alternative's bound (Corollary 2(ii)), although the alternative it favors is never entertained.

The signal's diagnostic power, large in aggregate, is fragmented across the menu, and the default remains defensible: fragmentation can produce a curiosity failure even when every alternative outperforms the default. Where crowding lowers the bounds without pushing them below threshold, alternatives are activated at smaller weights; since the information an agent seeks rises with the weight she places on an alternative (Proposition 2), search is weaker and the shift slower, so the joint failure of part (iii) is the limit of a graded effect rather than a knife edge. By the same mechanism, curiosity and the information acquired are non-monotone in the number of rivals: zero with none, positive with one that is activated, and tending to zero as rivals are added, with no information acquired beyond finitely many (Corollary OA.2 in the Online Appendix).

Interference effects have two direct applications to the design of information interventions.

Application 1: Targeting versus breadth. Consider an information intervention with a fixed amount of evidence to distribute across alternatives, so that each alternative's likelihood rises with the share it receives. Concentrating the budget on the most likely alternative drives its evidence cap to one as the budget grows, whereas spreading it uniformly leaves every cap below $\max_k L_k / \sum_{j \geq 1} L_j < 1$ at every budget, however large (Proposition OA.3 in the Online Appendix). Breadth fails not because it delivers less information but because spreading it raises every likelihood in the denominator of every bound, so the ratios barely move. Figure 2 shows the two paths for $K = 4$: the targeted bound passes 0.9 at a budget of about four, while the uniform bound stays below 0.29 at any budget. Enumerating many reasons a program might work therefore makes several alternatives moderately plausible and none plausible enough.

Application 2: Sequencing versus breadth. Suppose two alternatives must eventually be entertained. Under (4) the cost of holding both at weights w_1, w_2 is $\frac{\gamma}{2}(w_1^2 + w_2^2) + \eta w_1 w_2$, whereas two separate activation–signal–resolution cycles cost $\frac{\gamma}{2}(w_1^2 + w_2^2)$ in total: holding them simultaneously costs a stacking premium of exactly $\eta w_1 w_2$, and a larger hedging cost whenever the two models imply different posterior means, so absent discounting sequencing weakly dominates stacking (Proposition OA.1 in the Online Appendix). Sequencing also arises on its own. If one alternative clears its threshold, it is investigated first; if the evidence goes against it, every other bound rises, so a second alternative may clear its threshold in turn. If none clears its threshold, no bound moves, and the joint failure of Proposition 8(iii) persists until a new exogenous observation arrives (Corollary OA.3 in the Online Appendix). Endogenous exposure thus sharpens targeting: the decision maker can climb out of a fragmented menu on her own, eliminating hypotheses one at a time, but only after something has lifted one alternative over its threshold. Until then, the curiosity failure persists.

5 Creative Thinking: Compressing the Space of Mental Models

Sections 2–4 take the menu of mental models as fixed. This section and the next relax that assumption, and both are organized around one problem, *cognitive dimension reduction*. The space of possible mental models

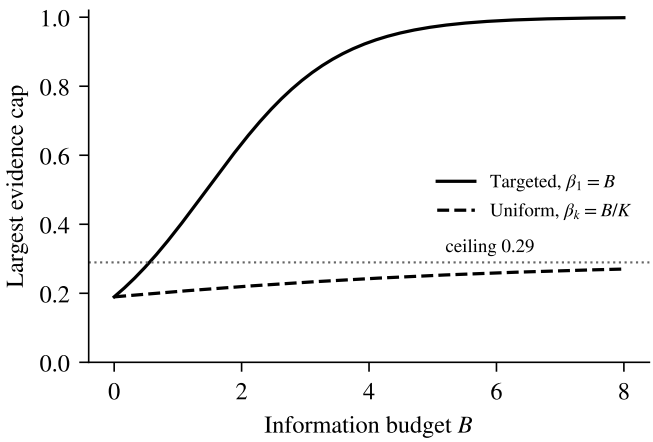


FIGURE 2. TARGETING VERSUS BREADTH IN INFORMATION INTERVENTIONS

Notes: The figure plots the largest evidence cap among $K = 4$ alternatives against an information budget B that multiplies each likelihood L_k by e^{β_k}, with $(L_0; L_1, \ldots, L_4) = (1.0; 0.55, 0.50, 0.45, 0.40)$. The targeted treatment puts the whole budget on the most likely alternative ($\beta_1 = B$); the uniform treatment spreads it evenly ($\beta_k = B/K$). The dotted line is the ceiling $\max_k L_k / \sum_j L_j \approx 0.29$ that no uniform budget exceeds (Proposition OA.3 in the Online Appendix).

is vast, and the results above show that a large menu defeats exploration. Every alternative that fits the evidence enters every evidence cap (Proposition 8), information spread across many alternatives caps the weight any of them can receive (Section 4.2), and even alternatives the agent would never entertain crowd out promising ones (Corollary 4). Holding several mental models at once is also superadditively costly (Section 2.4). An agent who represented every possibility as a separate mental model would rarely explore at all. Exploration requires keeping the effective menu small: describing many possible mental models with few reusable parts, and comparing few of them at a time. Creative thinking does this by compression, building new mental models from parts the agent already holds and pooling situations under common parameters.

Here, *creative thinking* is the construction of a mental model the agent has not used before from templates she already holds. A template is a reusable piece of explanation: an underlying cause together with the observable features it produces, such as rain, which shows up in wet soil and high yields, or pests, which show up in damaged leaves and poor yields. In the language of causal inference, a template is a mechanism, and a mental model is the causal graph assembled from such mechanisms and bound to a situation: a farmer who credits a neighbor's harvest to both the rain and the soil has combined two templates (Pearl, 2009; Behrens et al., 2018; Whittington et al., 2020). Creation starts from existing material: imagining a new experience depends on the memory system and works by recombining stored experience (Hassabis et al., 2007; Schacter and Addis, 2007; Gupta et al., 2010), while the templates themselves come from slower learning or from innate structure (Spelke and Kinzler, 2007). A library of M templates describes up to $2^M - 1$ mental models (Weitzman, 1998), so it represents a large space with few parts. The agent chooses steps rather than finished creations, since a creation cannot be ranked before it is conceived but a step can. Preferences decide whether a step is attempted, surprise gives it a direction, and evidence decides whether it is admitted (Campbell, 1960). The approach is close to combinational creativity and insight as a change of representation (Koestler, 1964; Boden, 2004; Knoblich et al., 1999), and differs from models of growing awareness, in which unforeseen options arrive from outside (Karni and Vierø, 2013). The model thus captures combinational and analogical creativity, the recombination and transfer of templates the agent already holds.[15]

[15] Boden (2004) distinguishes a third type, transformational creativity, which changes the space of possible ideas itself. On the neural view, imagination depends on the memory system and works by recombining stored experience (Hassabis et al., 2007; Schacter and Addis, 2007), and new structural representations are themselves learned from experience (Whittington et al., 2020). The model therefore reads transformational creativity not as creation from nothing but as recombination at the level of the templates themselves

5.1 Templates and mental models

Let $\mathscr{C}$ be a finite set of situations, such as plots, lessons, or problems, and let $\mathscr{T} = \{1, \ldots, M\}$ be a library of templates. Formally, a template is a causal mechanism: a latent cause together with the edges from it to the observable features it moves, as in a structural causal model (Pearl, 2009). I work with the linear Gaussian case, in which a template is a latent factor, a tuple $\tau_t = (b_t, \mu_t, \sigma_t^2, \rho_t)$. The loading vector $b_t \in \mathbb{R}^d$ records which observable features the cause moves: the direction in feature space along which it shows up, so that for rain it loads on soil moisture and yields. The prior $\mathscr{N}(\mu_t, \sigma_t^2)$ records how strongly the cause is expected to act, and its variance is the template's vagueness. The *generality* $\rho_t \in [0,1]$ sets how far that strength is invariant across situations. A *node* is a pair $(S, \mathscr{C}_S)$: a set of templates $S \subseteq \mathscr{T}$ and a scope $\mathscr{C}_S \subseteq \mathscr{C}$ of situations to which it is taken to apply. The node's generative process in situation $c \in \mathscr{C}_S$ is

$$x_c \mid \vartheta \;\sim\; \mathscr{N}\Big(\sum_{t \in S} \vartheta_{t,c}\, b_t,\; \Sigma\Big), \qquad \vartheta_{t,c} = \mu_t + \eta_t + \varepsilon_{t,c}, \qquad \eta_t \sim \mathscr{N}(0, \rho_t \sigma_t^2), \qquad \varepsilon_{t,c} \sim \mathscr{N}\big(0, (1-\rho_t)\sigma_t^2\big), \tag{18}$$

with all η_t and $\varepsilon_{t,c}$ independent of one another and of the noise. Templates contribute additively to what is observed, as causes do in a linear structural model. Each situation's parameter has the template's prior whatever its generality, and any two situations' parameters have correlation ρ_t, so a broader claim is not also a vaguer one. A template with $\rho_t = 0$ makes no claim across situations, since its parameters are independent draws from the prior. A template with $\rho_t = 1$ claims that one parameter governs every situation in the node's scope, and the set of situations over which its parameter is shared in this way is its *reach*: the situations across which the mechanism is invariant (Peters, Bühlmann, and Meinshausen, 2016), so that it transports from one to another (Pearl and Bareinboim, 2014). Intermediate values of ρ_t express partial pooling: the agent is unsure how far the mechanism transports. The mental models of Section 2.1 are the special case of one template and one situation: a prior $\mathscr{N}(\mu_k, \sigma_k^2)$ and a loading b_k. A node is thus a causal graph, as in models of causal narratives (Spiegler, 2016; Eliaz and Spiegler, 2020), assembled from reusable mechanisms. Holding a node other than the default costs mental effort, holding several at once carries the interference cost, and the weight on a node is capped by its evidence cap.

Definition 2 (Two types of creative thinking)**.** *Vertical creative thinking* adds a template to a node, $S \mapsto S \cup \{t\}$, for a given situation. *Horizontal creative thinking* extends a node's scope, $\mathscr{C}_S \mapsto \mathscr{C}_S \cup \{c'\}$, for given templates.

In causal terms, a vertical step adds a cause to the graph, and a horizontal step transports the graph to a new situation.

The two steps are the knowledge counterparts of the two kinds of innovation in endogenous growth. A vertical step is a step up a quality ladder (Aghion and Howitt, 1992): it explains a situation the agent already covers more deeply. A horizontal step resembles an expanding variety (Romer, 1990): it widens the range of situations an existing mental model serves. The two steps also reduce dimension in different directions. A vertical step adds structure within a situation and is judged as a comparison between two nodes, so the evidence is not divided among every assembly the library could form. A horizontal step pools situations: in the limit it replaces a free parameter for each situation with a single shared one.

Attempting a step to a node S' costs mental effort ζ, and the agent attempts it when the value of holding S', at the weight its evidence cap admits, covers the cost:

$$V\big(\overline{W}_{S'}(x), n^*\big) - V\big(\overline{W}_S(x), n^*\big) \;\geq\; \zeta. \tag{19}$$

or as the product of the slower learning by which new templates form. Every creation in the model is built from templates; that slower learning lies outside it.

5.2 Vertical creative thinking

Fix one situation and a node S. Write B_S for the matrix of the node's loadings, D_S for the diagonal matrix of their prior variances, and

$$V_S := \Sigma + B_S D_S B_S'$$

for the node's predictive covariance of the features. Let $r := x - B_S\mu_S$ be the part of the observation that node S did not predict, and let $h_t := b_t' V_S^{-1} b_t$ be the precision with which template t's direction can be read against everything node S already explains.

Proposition 9 (Justifying a step up)**.** *The Bayes factor of the node $S\cup\{t\}$ against S is*

$$\log \Lambda_{S\cup t:S} \;=\; \tfrac{1}{2}\Big[\, G_t - \log\big(1+\sigma_t^2 h_t\big)\Big], \qquad G_t := r' V_S^{-1} r - (r-b_t\mu_t)'\big(V_S + \sigma_t^2 b_t b_t'\big)^{-1}(r-b_t\mu_t), \tag{20}$$

where G_t is the gain in how well the residual is explained and $\log(1+\sigma_t^2 h_t)$ is the Occam penalty for the template's vagueness. If the template is centered, $\mu_t = 0$, then

$$G_t \;=\; \frac{\sigma_t^2 h_t}{1+\sigma_t^2 h_t}\, z_t^2, \qquad z_t := \frac{b_t' V_S^{-1} r}{\sqrt{h_t}},$$

where z_t is the standardized surprise of the residual along the template's direction. In a comparison of the two nodes, the step is justified, $\overline{W}_{S\cup t} > 1/2$, if and only if

$$z_t^2 \;>\; \frac{1+\sigma_t^2 h_t}{\sigma_t^2 h_t}\log\big(1+\sigma_t^2 h_t\big), \tag{21}$$

a threshold that is increasing in $\sigma_t^2 h_t$ and tends to one as $\sigma_t^2 h_t \to 0$.

Proposition 9 shows where compression pays. Because the agent evaluates a step, the comparison is between two nodes, and the test is one-dimensional: what matters is the surprise of the residual along the template's direction, z_t, net of an Occam penalty for the template's vagueness. The three margins of Section 3.3 reappear in a single direction: surprise and encoding both enter through z_t, the first through features S reads but mispredicts and the second through features S ignores, which cannot move any decision made under S but can justify a new template (Corollary OA.4 in the Online Appendix), while specificity is the Occam penalty. Compression has a price. If the agent tries many templates after seeing x and keeps the best, the candidates she tried form a menu in all but name, and the crowding that would have disciplined that menu is lost.

5.3 Horizontal creative thinking

Now hold the templates fixed and ask whether a node applies across N situations. Consider one template, with prior $\mathcal{N}(\mu,\sigma^2)$ and generality ρ; in a node with several templates the argument applies to each template's parameter given the others. Let s_c be the template's sufficient statistic in situation c, read as in Lemma 1, with $s_c \mid \vartheta_c \sim \mathcal{N}(\vartheta_c, v)$. The *local* node sets $\rho = 0$ and treats the N parameters as unrelated draws from the prior. The *general* node has $\rho > 0$, as in (18). Write $d_c := s_c - \mu$ for each situation's departure from the prior, $\bar{d} := N^{-1}\sum_c d_c$ for the departure the situations share, and $SS := \sum_c (d_c - \bar{d})^2$ for their dispersion.

Proposition 10 (Justifying transfer)**.** *Let $a := v + (1-\rho)\sigma^2$ and $g := v + \sigma^2 + (N-1)\rho\sigma^2$. The Bayes factor*

of the general node against the local node is

$$\log \Lambda_{\text{gen:loc}} = \underbrace{\tfrac{1}{2}\Big[N\log(v+\sigma^2) - (N-1)\log a - \log g\Big]}_{\textit{Occam gain: a shared parameter for N situations}} - \tfrac{1}{2}SS\Big[\frac{1}{a} - \frac{1}{v+\sigma^2}\Big] + \tfrac{N}{2}\bar{d}^2\Big[\frac{1}{v+\sigma^2} - \frac{1}{g}\Big]. \tag{22}$$

It is zero when $N = 1$. For $N \geq 2$ and $\rho > 0$, it is increasing in the shared departure $\bar{d}^2$ and decreasing in the dispersion SS, and the general node is justified if and only if SS falls below a threshold that is increasing in $\bar{d}^2$. At $SS = \bar{d} = 0$ it is increasing in ρ. At $\rho = 1$ the Occam gain is increasing in N.

Horizontal creative thinking reduces dimension across situations. The local node carries N free parameters; the general node at $\rho = 1$ carries one, and the Occam gain from that reduction grows with N. Evidence from different situations then bears on a single parameter rather than being divided among N, the cross-situation counterpart of concentrating information on one alternative (Section 4.2).

5.4 Deeper explanations apply more widely

A *mechanism template* is invariant across the situations in which the mechanism operates: its generality is $\rho_t = 1$ on its reach R_t, and outside R_t it does not contribute to what is observed.

Proposition 11 (Vertical mechanisms determine the extent of horizontal transfer)**.** *Let $S' = S \cup \{t\}$ add a mechanism template t whose parameter is common to a set of situations $R_t \subseteq \mathscr{C}$. Then:*

(i) The reach of S' is R_t: the parameter of t is shared on R_t, and outside R_t the node is as before, so the reach is a property of the template rather than a choice.

(ii) For $N \geq 2$ situations in R_t whose evidence agrees ($SS = \bar{d} = 0$), the transfer Bayes factor (22) *is strictly larger under S' than under any node with $\rho < 1$.*

(iii) For situations whose evidence disagrees, the penalty on dispersion, $\frac{1}{2}SS[1/a - 1/(v+\sigma^2)]$, is increasing in ρ and largest at $\rho = 1$.

Mechanisms are the limit of compression: one parameter for every situation in which the mechanism operates, with a reach set by the template rather than chosen. The property that lets agreeing evidence pool (part (ii)) also makes disagreeing evidence count most heavily against the mechanism (part (iii)). Maximal compression is maximal exposure to refutation, which is why compression needs the discipline of the next section. In the language of growth theory, a mechanism template is a general-purpose technology (Helpman and Trajtenberg, 1998): one step up the ladder raises explanatory quality in every situation where the mechanism operates, and by part (ii) it raises the return to expanding variety over the mechanism's reach. Vertical and horizontal creative thinking are therefore linked, unlike in models that treat the two kinds of innovation as separate activities (Aghion and Howitt, 1992; Howitt, 1999). Because creation draws only on the templates the agent holds, creative capability can be cultivated: providing templates, and above all mechanisms, enlarges both the set of mental models she can construct and the range of situations to which they transfer. This matters for training teachers and agricultural extension officers: a set protocol gives them a precise local mental model, which pays a small Occam penalty and is easily activated where it was designed but carries no claim beyond it; understanding why a practice works supplies a mechanism template, whose gain is zero in a single setting and grows with the number of classrooms and farms it covers (Proposition 10), at the cost of greater exposure to evidence that disagrees (Proposition 11(iii)). The template does not deliver the adaptations ready-made: each is a creative step the trainee must take and test herself, so the payoff relies on her continued creativity and curiosity.

6 Critical Thinking: Pruning the Menu with Fresh Evidence

Compression creates its own risk. A library of templates makes candidate mental models cheap to construct, and an agent who searches them after seeing the evidence and keeps the best fit has considered a large menu while counting only the winner. The rivals she discarded drop out of the evidence cap, and with them the crowding of Proposition 8 that a menu of that size would have faced. Critical thinking prunes the menu, admitting only mental models that survive evidence not used to form them.

The problem is familiar to empirical economists as multiple hypothesis testing. A researcher who tests many outcomes and reports the one that is significant finds false positives at a rate that grows with the number of tests, and flexible analysis can make almost any result look significant (Simmons et al., 2011; Gelman and Loken, 2014). Corrections for multiple inference and pre-analysis plans are now standard in field experiments for this reason (Anderson, 2008; Olken, 2015; List, Shaikh, and Xu, 2019). Critical thinking is the cognitive counterpart: a person who searches many explanations after the fact faces the same multiplicity, and the same remedies apply.

Coming up with a new explanation is only half of learning. The other half is judging whether it is right. A common failure is to pick an explanation because it fits and then count that same fit as evidence that it is right: the data are used twice. A farmer whose neighbor harvests far more than usual runs through possible reasons, such as the seed, the rain, the planting date, and the spacing, and settles on spacing because it matches what she saw most closely. With enough candidate reasons, one is bound to match by chance, so the match alone says little about whether spacing caused the harvest. A critical thinker asks whether she would have found a fitting story whatever had happened, and she checks the story on cases she did not use to find it. Reading meaning into chance coincidences is also how superstitions form (Skinner, 1948). Here, *critical thinking* is the disposition to distinguish predictions from after-the-fact explanations, so that every mental model, new or old, is judged only by evidence that was not used to form it. It is close to the disposition to evaluate arguments independently of prior belief (Stanovich and West, 1997) and opposite to confirmation bias (Nickerson, 1998). This is one component of critical thinking, the discipline against after-the-fact fitting; the analysis of arguments and the evaluation of sources lie outside the model.

6.1 The search discount

Fix a node S and an observation x. After seeing x the agent searches a class $\mathscr{S} \subseteq \mathscr{T}$ of candidate templates and keeps $t^* := \arg\max_{t \in \mathscr{S}} \Lambda_{S \cup t:S}(x)$, with the Bayes factor of Proposition 9. A Bayesian who places equal prior weight on each candidate assigns prior odds $1/|\mathscr{S}|$ to t^*.

Definition 3 (Critical thinking)**.** An agent with critical thinking $\kappa \in [0,1]$ evaluates a node constructed by searching a class $\mathscr{S}$ after observing x at the log odds

$$\log \Lambda_{S \cup t^*:S}(x) \;-\; \kappa \log |\mathscr{S}|. \tag{23}$$

A model specified before x, such as one supplied by teaching, has $|\mathscr{S}| = 1$ and is not discounted, and neither is evidence that was not used in the search.

With $\kappa = 1$ the candidates searched share equally the prior weight that a single alternative named in advance would carry, so the winner is charged for the rivals the search considered. The discount depends on the breadth of the search, not on its outcome, and plays the role of a multiplicity correction: an explanation named before the evidence is treated like a hypothesis registered in a pre-analysis plan, while one found by searching many candidates is held to a higher bar.

6.2 Search breadth and false insights

A *false insight* is a template admitted when the observation was generated by S. Let the candidates be centered, $\mu_t = 0$, read with common precision $y := \sigma_t^2 h_t$, and read directions that are orthogonal in the metric V_S^{-1}. Admission requires that the evidence cap of $S \cup \{t^*\}$ in its comparison with S exceed one half.

Proposition 12 (Critical thinking and breadth)**.** *Under these conditions the agent admits a template if and only if*

$$\max_{t \in \mathscr{S}} z_t^2 \;>\; c_\kappa(y, |\mathscr{S}|) \;:=\; \frac{1+y}{y}\Big[\log(1+y) + 2\kappa \log|\mathscr{S}|\Big], \tag{24}$$

and the probability of a false insight, $\pi_\kappa(|\mathscr{S}|)$, satisfies:

(i) If $\kappa < y/(1+y)$, then $\pi_\kappa(|\mathscr{S}|) \to 1$ as $|\mathscr{S}| \to \infty$. In particular a naive agent forms a false insight with probability approaching one when she searches broadly.

(ii) If $\kappa > y/(1+y)$, then $\pi_\kappa(|\mathscr{S}|) \le (1+y)^{-(1+y)/(2y)}\,|\mathscr{S}|^{\,1-\kappa(1+y)/y} \to 0$. This bound does not require independence across candidates.

(iii) For a candidate that is right, the surprise required for admission rises with the breadth of the search, by $2\kappa\frac{1+y}{y}\log|\mathscr{S}|$ in z_t^2.

The bar in (24) rises in proportion to $\log|\mathscr{S}|$, the rate at which a Bonferroni critical value for $|\mathscr{S}|$ normal tests rises, so breadth of search and discipline in judging it go together. Part (iii) is the potential pitfall: critical thinking also demands more of a correct insight. The discount matters beyond single steps. Each false insight a naive agent admits is a mental model that fits the evidence, so it enters the denominator of every evidence cap and dilutes the support for the alternative that is right (Proposition 8), whether or not she would ever entertain it (Corollary 4). By keeping chance fits off the menu, critical thinking can make the right alternative easier to activate. The same discount applies to old mental models: critical thinking loosens a default that owes more to after-the-fact search than the alternative does, so it is not a bias toward the familiar (Proposition OA.4 in the Online Appendix).

6.3 Fresh evidence and the pitfall of critical thinking

Proposition 13 (Fresh evidence)**.** *Let t^* be chosen on x, and let $x'_1, \ldots, x'_m$ be later observations independent of x given the process that generates them, such as observations from new plots. Then the critical thinker's log odds for $S \cup \{t^*\}$ against S are*

$$\log \Lambda_{S \cup t^* : S}(x) - \kappa \log|\mathscr{S}| + \sum_{i=1}^{m} \log \Lambda_{S \cup t^* : S}(x'_i),$$

so they differ from the naive agent's by the constant $\kappa \log|\mathscr{S}|$. Each fresh observation changes the log odds in expectation by the amount by which $S \cup \{t^*\}$ fits it better than S, in Kullback–Leibler divergence from the process that generates it, as in Corollary 3. As $m \to \infty$, the bounds in both agents' evidence constraints for $S \cup \{t^*\}$ converge to one if t^* improves the fit and to zero if it worsens it.

Fresh evidence plays the role of a replication or hold-out sample, and it is how the menu is pruned over time. A node whose predictions fail on observations not used to form it sees its evidence cap fall toward zero, while the discount, charged once, matters most when evidence is scarce. Extending a mental model to a new situation, the horizontal step of Section 5, is therefore also an out-of-sample test.

Fresh evidence arrives, however, only if it is collected, and the agent collects evidence only through the mental models she holds. Say that an observation *bears on* a template t if its distribution differs under $S \cup \{t\}$ and under S. Critical thinking can therefore produce a curiosity failure for a correct insight: when such

observations are sought only by an agent who places weight on $S \cup \{t\}$, the discount that keeps the template off the menu also keeps the agent from collecting the evidence that would admit it, and her log odds stay where the discount left them, whether or not the template is correct (Corollary OA.5 in the Online Appendix). The higher bar of Proposition 12(iii) is temporary only when evidence bearing on the template arrives without being sought, as when a neighbor's harvest is visible anyway or a program delivers the evidence, in which case Proposition 13 applies.

Creative thinking can offset the discount, but only in particular forms. The discount grows with the number of candidates searched, so generating more candidates after seeing the evidence raises it. What lowers the surprise a correct template needs is a narrower search, a template held before the evidence, or a template whose parameter is common to many situations, so that the evidence from all of them bears on it at once.

Corollary 5 (Creative thinking and the discount)**.** *Under the conditions of Proposition 12, let x contain observations from N situations that are independent given each node and share the node's predictive covariance V_S, and let t be a centered mechanism template whose parameter is common to the N situations, with precision y in each. Let $z_{t,c}$ be the standardized surprise along t in situation c, and $Z_t := N^{-1/2}\sum_c z_{t,c}$. Then:*

(i) The critical thinker admits t if and only if $Z_t^2 > c_\kappa(Ny, |\mathscr{S}|)$.

(ii) The additional surprise critical thinking requires, $c_\kappa(Ny, |\mathscr{S}|) - c_0(Ny, |\mathscr{S}|) = 2\kappa \frac{1+Ny}{Ny} \log |\mathscr{S}|$, is zero when $|\mathscr{S}| = 1$, increasing in $|\mathscr{S}|$, and decreasing in N.

(iii) If t is correct, with parameter $\vartheta_t \neq 0$, the probability that the critical thinker admits it tends to one as $N \to \infty$, for every $\kappa \in [0,1]$ and every $|\mathscr{S}|$.

Part (ii) makes the three routes precise: the discount vanishes for a template held before the evidence, falls with a narrower search, and falls as a template applies to more situations. Part (iii) shows that a mechanism template applied widely enough is eventually admitted however strong the discount. Pooling has a second effect: by Proposition 12 with precision Ny, the threshold value of κ that keeps false insights out rises to $Ny/(1+Ny)$. A full discount, $\kappa = 1$, still meets it, so an agent who discounts fully and applies templates widely both keeps chance fits off the menu and admits the correct template. Critical thinking thus raises the return to creative thinking that narrows the search or pools evidence across situations, and lowers the return to creative thinking that only adds candidates.

7 Concluding Remarks

This paper asks what curiosity is: what leads a person to seek the information required to explore new perspectives. It treats contextual shifts in mindsets as a choice of how much model uncertainty to entertain. Associative memory supplies a familiar default. After an initial signal, an agent with latent, less precise alternatives jointly chooses how much weight to place on each and what information to acquire, and how much, disciplined by an evidence constraint that caps the weight on any alternative at its Bayesian posterior. Curiosity then arises endogenously as the value of resolving the model uncertainty the agent chooses to entertain.

Because agents can seek information and reason beyond what associative memory supplies, exploration is both triggered and tested by evidence, which revises hallmark behavioral effects in associative-memory models. Contextual cues payoff-irrelevant under the default sway decisions only when they activate an alternative under which they are relevant, and the curiosity this stimulates tests that alternative, so the shift lasts only if evidence supports it. Yet rival explanations can stifle exploration of a promising alternative by dividing the evidence, even if the agent would never entertain them. Sections 5 and 6 relax the assumption of a fixed menu and treat creative and critical thinking as cognitive dimension reduction: creative thinking compresses the space of mental models into a few reusable templates, and critical thinking prunes the menu to those that survive evidence not used to form them.

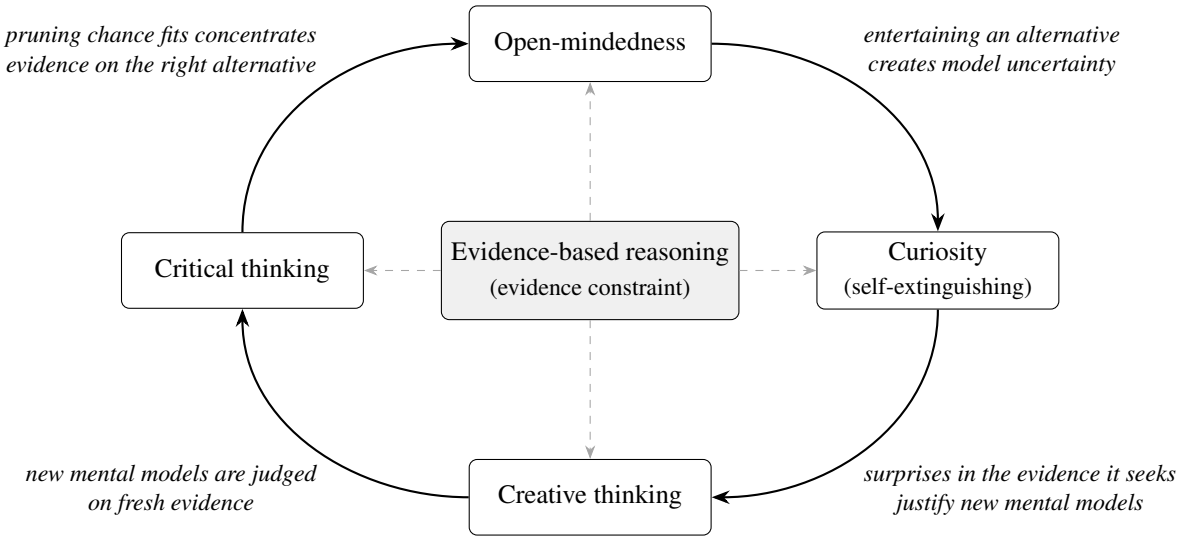


FIGURE 3. THE CYCLE OF OPEN-MINDEDNESS, CURIOSITY, CREATIVE THINKING, AND CRITICAL THINKING

Notes: Each solid arrow summarizes a result. Entertaining an alternative within its evidence cap creates the model uncertainty that curiosity values; the surprises in the evidence curiosity seeks justify new mental models; new mental models are judged on evidence not used to form them; and keeping chance fits off the menu concentrates the evidence on the right alternative. Curiosity extinguishes itself as the evidence it gathers resolves model uncertainty, so the cycle continues only if creative thinking replenishes it. The dashed arrows indicate that every process is disciplined by the evidence constraint.

Taken together, the results suggest that open-mindedness, curiosity, creative thinking, and critical thinking form a cycle within the brain's evidence-based reasoning (Figure 3). Open-mindedness, entertaining an alternative to the extent the evidence allows, is what curiosity requires: curiosity is the value of resolving model uncertainty, and there is none to resolve until an alternative is entertained. Curiosity tends to extinguish itself, since the evidence it gathers resolves the uncertainty that prompted it: the mental models come to agree or one of them prevails. Creative thinking can replenish that uncertainty. The observations curiosity seeks were not used to form the mental models, so their surprises along a template can justify a new mental model, and they are the fresh tests critical thinking requires. Critical thinking keeps chance fits off the menu, where they would dilute the evidence cap of the right alternative. Its potential pitfall ties the four together: it raises the bar for a correct template more the broader the search that found it, which creative thinking that narrows the search or pools evidence across situations offsets, and it can leave a correct template untested unless curiosity, or evidence that arrives without being sought, supplies fresh evidence. Finally, a mental model that sets the cycle going in one situation does not carry over for free: extending it to new situations is a horizontal step that the evidence must justify and that costs mental effort.

The framework points to an integrative approach to education and training, and the stepping stones in Appendix B show how its two parts fit together. The "Learning to Learn" approach to teacher training and community transformation acts on the menu (Nourani et al., 2025, 2026). Reflection on inquiry-based against traditional instruction, together with simulation and discussion with peers, raises the evidence caps of a class of inquiry-based pedagogies, both by directing attention to features of the classroom the traditional model ignores and by sharpening the templates behind them rather than giving out recipes; drawing analogies trains the horizontal steps that carry a template to new situations; and separating facts from opinions includes the discipline against after-the-fact fitting formalized here. School-based agricultural education acts on the information environment, as a platform that offers experiential science learning opportunities and innovative farming techniques (Lee, 2026): school farms serve as demonstration plots, science lessons and student home projects make evidence cheap and draw it from many plots at once, and extension officers return with feedback from time to time. Each part of one program pairs with a part of the other. Sharpened mental models are investigated only if signals are cheap enough, and a higher evidence cap raises the information acquired

more the cheaper the signals, while information that no mental model reads goes unused (Hanna et al., 2014). Critical thinking can leave a correct explanation untested unless evidence arrives without being sought, which demonstration plots and extension visits supply. And analogies that carry a template across plots clear the discount only with evidence from many situations, which home projects supply. Together the two programs let teachers and students create their own mental models one at a time, from an inquiry routine made familiar through practice, with lower but more certain payoffs, through tests that either eliminate it or justify adding a template, to mental models with higher but less certain payoffs. My ongoing research pairing the two programs is designed to test this integration.

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A Relation to Neuroscience and Existing Models

A.1 The neuroscientific regularities the model reproduces

The paper's claim is that a well-established neuroscientific theory disciplines the decision problem, and that claim is testable against the theory's own findings. Table OA1 in the Appendix matches nine documented regularities to the model feature that produces each. Three correspondences are close. The evidence cap is, term for term, the posterior over contexts that Sanders et al. (2020) argue governs remapping, so the model's central constraint is not an analogy but the same object placed in a decision problem. The two regimes of Proposition 1 reproduce rate and global remapping (Leutgeb et al., 2005a,b; Wills et al., 2005), and the model supplies what the neuroscience does not have, a classifying variable: the disagreement between models about what to do relative to the cost of effort. And the interference term reproduces the working-memory capacity limit (Cowan, 2001; Bays and Husain, 2008) in the only form a decision model can use, a superadditive cost.

Two are looser. Flickering between mental models (Jezek et al., 2011) is a within-trial phenomenon on a timescale of seconds, and the model's interior weights are its natural counterpart but have no timescale; and global remapping replaces one code by another, whereas the abrupt regime moves the weight to the evidentiary limit $\overline{W}_k < 1$. On three matters the model is silent: the geometry of the code, the automatic assignment itself, which it takes as given in (7), and whether the variable that sorts an animal into rate or global remapping is the model's γ against $2D_k$. The last is the honest gap, since those experiments manipulate environmental similarity rather than disagreement about a payoff.

A.2 What distinguishes the model

Table OA2 in the Appendix lists the phenomena delivered above and marks which of the closest alternatives also deliver them: Bayesian model averaging, hypothesis testing in the sense of Ortoleva (2012), model persuasion in the sense of Schwartzstein and Sunderam (2021), optimal expectations (Brunnermeier and Parker, 2005), and associative-memory categorization in the sense of Bordalo et al. (2026). Two rows carry most of the weight. The one-sidedness of the constraint is what separates the model from Bayesian model averaging: a Bayesian is moved by evidence exactly as much as Bayes' rule says, whereas the agent here may stay with the default against the evidence but cannot run ahead of it, which is why the default is sticky and why interventions must clear a threshold. And the activation threshold is what separates it from hypothesis testing and model persuasion: those frameworks have the agent switch when the evidence is decisive, but they have no counterpart to holding an alternative at a small weight, investigating it or not, and being crowded by other alternatives, which is where the model's design content lives.

Behavioral puzzles. Several regularities in belief updating and information acquisition, usually explained separately, follow from the same structure. Underreaction to evidence (Edwards, 1968; Benjamin, 2019) is the one-sidedness of the constraint: below the activation threshold the weight stays at zero although the Bayesian bound is positive. That underreaction coexists with abrupt revisions is the all-or-nothing regime of Proposition 1, in which the same agent ignores weak evidence and jumps on strong evidence. The avoidance of freely available, decision-relevant information (Golman et al., 2017) needs no taste for ignorance: below the threshold there is no second model on which the information would bear, so it has no value (Corollary 1). The failure to notice features that matter (Schwartzstein, 2014; Hanna et al., 2014) is the attention mapping of the model in use (Lemma 1), which changes, and with it what is noticed, only when weight moves. And curiosity that peaks at intermediate confidence (Loewenstein, 1994) is the inverted U of Definition 1 in the weight.

B Mental Models with More Certain Payoffs as Stepping Stones

The results in Sections 3–6 concern a single step. Taken over repeated stages, with the evidence caps carried forward and the agent holding one alternative at a time, they describe a path.

Corollary B.1 (Stepping stones)**.** *Let Assumption 1 hold, let φ be linear, and let signals cost c.*

(i) (The first step.) *An alternative k is investigated, $w_k^* > 0$ and $n^* > 0$, only if $\overline{W}_k > 2\lambda\nu_y^2 c P_k(0)^2$. When the evidence constraint binds, $n^* = \nu_y^2\left[(\overline{W}_k/(2\lambda\nu_y^2 c))^{1/2} - P_k(0)\right]^+$, which is increasing in $\overline{W}_k$ and decreasing in c, and wherever $n^* > 0$ the effect of a higher cap on the information acquired is larger the lower the cost of signals, $\partial^2 n^*/\partial\overline{W}_k\,\partial c < 0$.*

(ii) (Elimination.) *If k is investigated, the signals are generated by another mental model, and they discriminate between the two, then $\Lambda_k \to 0$ almost surely as signals accumulate, and the bound in the evidence constraint of every rival $j \neq k$ rises to $\Lambda_j/(1+\sum_{i\neq k}\Lambda_i)$, so that a rival blocked by its cap at the start can be activated.*

(iii) (Building.) *If k is a node S and the observations acquired while investigating it bear on a template t, the vertical step to $S\cup\{t\}$ is admitted if and only if the surprise along t exceeds the bar $c_\kappa(y,|\mathscr{S}|)$ of Proposition 12, where $|\mathscr{S}|$ is the number of templates considered; when t is the only one, $|\mathscr{S}| = 1$ and the bar is that of Proposition 9.*

A mental model's prior becomes precise through use, as the default's has, or through practice with the templates it is built from. Sharpening a class of templates raises the evidence caps of the mental models assembled from them together, since a more precise template pays a smaller Occam penalty. Rivals still divide the evidence (Proposition 8), so the first alternative investigated is the one whose cap is highest, typically the most familiar among those that fit the evidence. It may promise little, since a precise prior leaves little to learn and so carries a smaller value gap (Section 3.3). Testing it advances the agent by part (ii) if it is wrong and by part (iii) if it is right but incomplete, and each step searches narrowly, so critical thinking discounts it little. At each step the agent creates the next mental model herself, from the surprises her own tests uncover, rather than adopting a finished one. A recipe, by contrast, is a finished mental model: precise enough to be activated where it was designed, but it supplies no template from which to build further (Section 5.4). Mental models with lower but more certain payoffs thus serve as stepping stones to ones with higher but less certain payoffs, whose priors are vague because the agent has yet to use them. Two conditions matter: sharpening raises a cap only if the sharpened model fits what is observed, and building requires that the observations gathered while testing one mental model bear on the next template, which by Proposition 6 is not automatic. Part (i) also identifies two levers that are complements: raising the evidence caps of the alternatives, and lowering the cost of signals that bear on them.